

Bachelor in **Applied Data Science**

DEVELOPING ML MODEL FOR EARLY DETECTION OF MPOX USING DIGITAL SKIN IMAGES

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Abstract

Monkeypox (Mpox) appear similar too ther skin diseases while presenting diagnostic challenges. This study develops and evaluates deep learning models for the early detection for monkeypox classification using digital skin lesion images and uses the publicly available Monkeypox Skin Image Dataset to develop and evaluate multiple convolutional neural network (CNN) architectures, including ResNet50, DenseNet169, EfficientNetB7, InceptionV3, VGG16, MobileNetV2, Xception and ensemble methods, for identifying skin illnesses as monkeypox, chickenpox, measles, and normal skin.

A comprehensive machine learning pipeline has been developed with including transfer learning, data augmentation, stratified k-fold cross-validation, and class weighting . The top performing individual model ResNet50, attained an accuracy of 96.9% and an F1-score of 96.0%, but an ensemble method enhanced performance to 98.0% accuracy. Grad-CAM visualizations were employed to improve model explainability, allowing for the validation that classifications have a basis in clinically pertinent image features. Additional the final model was implemented via a Gradio interface, providing an accessible interface for clinical use. This study highlights the viability of CNN based approaches for early detection of monkeypox, providing an opportunity for more accessible and efficient diagnosis in environments with limited resources.

Keywords: *Monkeypox, Deep learning, Convolutional neural networks, Pre-trained, Transfer learning, Ensemble , Explainable, Gradio*

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1

Introduction

1.1 Introduction

Monkeypox, also known as Mpox, is a viral infectious disease found in the genus Orthopoxvirus of the family Poxviridae, along with the variola virus, the cause of smallpox (World Health Organization, 2024).

Monkeypox was first found in monkeys in Denmark in 1958 and in humans in 1970 in the Democratic Republic of Congo. Even after being a neglected tropical disease for many years, it is now becoming a major global threat. Earlier, there have been many outbreaks in non-endemic countries, including Europe and the USA, which has made the development of new and highly effective diagnostic methods all the more important. The presence of people who are in apparent good health but have the virus makes the situation worse by causing difficulty with public health response. The disease manifests with symptoms similar to smallpox, such as fever, lymphadenopathy, and characteristic rash, although it is typically less severe. The diagnosis must be made quickly and accurately, and the absence of licensed vaccines also makes testing essential. The most common techniques (Figure 1.1) used to diagnose Monkeypox are polymerase chain reaction, also known as PCR, electron microscopy, and serological techniques (Stefano et al., 2023).

The price and requirements needed to perform this test limit its use in resource-limited areas. Electron microscopy provides a visual image of the Monkeypox virus with technical equipment, and is used mainly for research purposes. Serologic testing methods include enzyme-linked immunosorbent assays (ELISAs), which can be used to detect the presence of antibodies against the virus, but there is no quick way to tell whether the infection is current or if it has been in the past (Stefano et al., 2023).

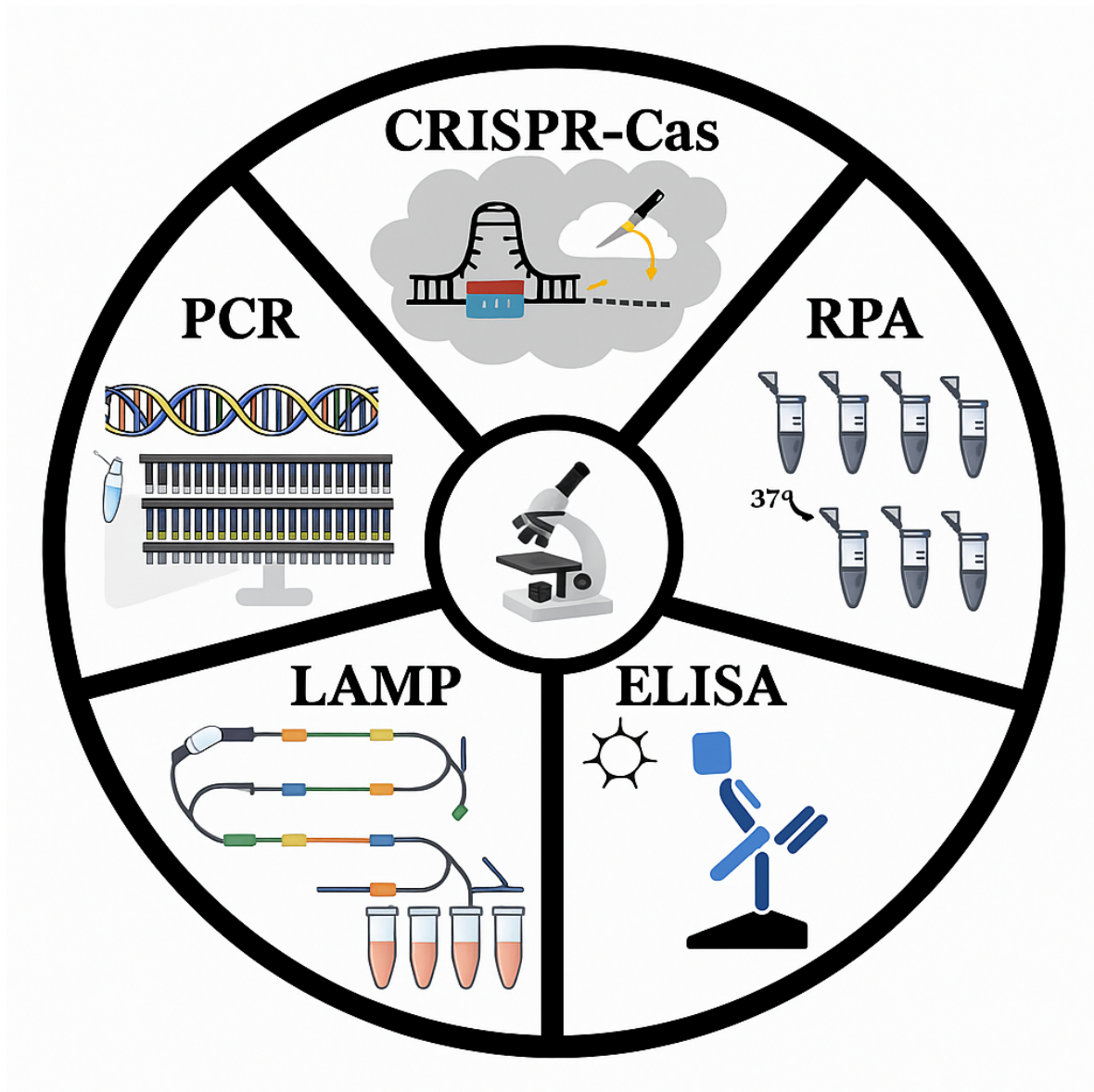


Figure 1.1: Main methods used to detect the monkeypox virus.

Unfortunately, diagnostic methods such as polymerase chain reaction (PCR) and serology assays, although reliable, require expensive laboratory setups and multiplex requiring skilled workers that are not available in many low resource settings. As a result, there is an urgent need to explore alternative diagnostic solutions that can be widely deployed, are cost effective, and can provide timely results to help curb the spread of monkeypox. (World Health Organization, 2024).

1.2 Background and Related Work

The recent revival of infectious diseases such as monkeypox emphasizes the urgent necessity of fast and precise diagnosis instruments. Emphasizing the need of automated solutions, traditional diagnostic approaches may find it difficult to differentiate monkeypox from other visually similar diseases such as chickenpox and measles. Particularly Convolutional Neural Networks (CNNs), deep learning

(DL) has become a transforming tool for medical imaging with strong features extraction and disease classification (O'Shea & Nash, 2015)

However, training CNNs from scratch is much challenged by the lack of annotated medical datasets. Transfer learning using ImageNet or another pre trained model has shown promise in tackling this problem. Transfer learning improves generalization and lowers computing needs through using features learnt from big datasets (Deng et al., 2009).

For example, pre trained, models like ResNet and EfficientNet have been fine-tuned to highly accurately classify diseases such monkeypox. By increasing variability as well as improving model performance, data augmentation methods such rotation and flipping help to further decrease dataset limitations (Ahsan et al., 2022).

In addition to these studies Transfer learning and CNNs have been successfully used in several studies to identify monkeypox. By means of ResNet50 and MobileNetV2, attained 93–99% accuracy, therefore proving the value of pre-trained models and data augmentation (Ahsan et al., 2022). In a similar created MonkeyNet, a DenseNet 201-based architecture, with 98.91% accuracy on enhanced datasets (Bala et al., 2023). Maqsood presented MOX-NET, which combined features from many CNN architectures to classify monkeypox, chickenpox, measles, and normal skin with 98.64% accuracy (Maqsood et al., 2024).

Through combining predictions from several models, hybrid and ensemble methods improve diagnostic accuracy even more. To attain 93.39% accuracy (Pramanik et al., 2023), blended predictions from InceptionV3, DenseNet169, and Xception). Attaching an 98.7% accuracy on the Monkeypox Skin Images Dataset (Haque et al., 2023) also used an ensemble model with DenseNet121, ResNet50, and Inception V3.

Moreover, the interpretability of Deep learning models in medical applications have become much more interpretable in great part to explainability strategies such Grad CAM. These instruments create visual heatmaps showing areas of interest in input photos, verifying forecasts, and so building clinician confidence. For monkeypox detection, Grad-CAM has been applied successfully with models including DenseNet 201 and MobileNetV3 (Altun et al., 2023; Selvaraju et al., 2016).

Finally, (Altun et al., 2023) investigated transfer learning to address the challenge of small data samples in training CNN-based models. They showed that pre-trained models and transfer learning methods significantly increased monkeypox detection performance. Furthermore (Ahsan et al., 2023) explored deep transfer learning to facilitate model adaptability to new data, which is crucial in outbreak scenarios.

In conclusion, the diagnosis of monkeypox and other similarly related diseases, the use of deep learning especially CNNs and transfer learning has shown to be a transforming approach. Studies have shown the success of using pre trained models, data augmentation techniques, and ensemble methods to get high diagnosis accuracy despite challenges including limited annotated datasets. Furthermore, the use of explainability tools such as Grad CAM has improved these models interpretability in application. Focusing on scalable solutions and flexibility to new data will be crucial as developments go forward to help to control future outbreaks and improve diagnostic capacity all around

1.3 Problem Statement

Developing a system to assist medical professionals in the early detection of disease, reducing human subjectivity, and enhancing diagnostic efficiency is the challenge of diagnosing monkeypox through digital skin images with advanced machine learning models. In particular, pretrained convolutional neural networks (CNNs) and transfer learning have enabled the field of medical imaging to make significant progress in the diagnosis of maladies using medical images that is effective because manual image analysis is a time consuming and errorprone procedure, and these models assist in overcoming these.

1.4 Research Objectives

The primary objective of this research is to develop a robust and accurate deep learning model for the early detection and classification of monkeypox using skin lesion images.

In order to achieve this primary objective, the following sub-objectives have been identified:

1. Evaluate and compare different pre-trained CNN architectures for monkeypox detection and apply transfer learning to improve the performance of CNN models on limited datasets.
2. Develop a method to differentiate between monkeypox and other diseases similar to mpox.
3. Study and integrate explainability techniques to understand and interpret model decisions and implement model generalization to ensure effectiveness.
4. Validate the performance of the developed models using standard metrics such as accuracy, sensitivity, and specificity.
5. Explore the potential integration of the developed model into an application (Windows or web interface) for real time detection and diagnosis.

1.5 Research Methodology

The research will follow a systematic approach aligned with the stated objectives:

1.5.1 Data Collection and Preprocessing

- **Data Collection:** Using public medical image databases to gather a diverse set of annotated images of monkeypox and other skin condition.
- **Data Preprocessing:** Apply image enhancement techniques such as normalization, noise reduction, and segmentation to improve image quality (Nayak et al., 2023).
- **Data Augmentation:** Implement techniques like rotation, flipping, scaling, and brightness adjustments to increase dataset diversity and handle class imbalance (Altun et al., 2023).

1.5.2 Model Development Using CNNs

- **Model Selection:** Utilize multiple pre-trained CNN architectures known for their effectiveness in image classification tasks (ResNet, EfficientNet,DenseNet).
- **Transfer Learning:** Leveraging pre-trained weights from ImageNet
- **Training Strategy:** Two stage approach with initial frozen base model training followed by fine-tuning of deeper layers

1.5.3 The Model Generalization

Using data augmentation, transfer learning and cross-validation to ensure that the model works effectively on data. Accuracy and adaptability will be improved through explainable approaches and ongoing updates.

1.5.4 Integration of Explainable

- **Explainability Methods:** Incorporate methods like Grad-CAM to highlight the areas of the image that contribute most to the model's predictions (Haque et al., 2023).

1.5.5 Model Evaluation and Validation

- **Performance Metrics:** Evaluate the model using metrics such as accuracy,Recall, precision, F1-score, and area under the ROC curve (Ahsan et al., 2023).
- **Visualization:** Confusion matrices to analyze classification patterns
- Medical doctors will be involved in validating the model's performance and ensuring its clinical applicability.

1.5.6 Deployment Framework Development

- **Deployment Framework:** Create a user-friendly interface, possibly a simplified Windows app or web interface. (Alhasson et al., 2023).

1.6 Primary Data Set

Two publicly available datasets on Kaggle: [Monkeypox Skin Image Dataset](#) the includes images of various skin conditions, including monkeypox. Preprocess the data and train machine learning models for early detection. Data are anonymized and made public to reduce ethical concerns.

1.7 Scope and Limits

Scope:

- Development and evaluation of CNN-based models for monkeypox detection.
- Utilization of transfer learning to enhance model accuracy with limited data (Ahsan et al., 2023; Altun et al., 2023).
- Comparison of model performance with existing diagnostic methods

Limits:

- Development of diagnostic tools for diseases other than monkeypox.
- Analysis of non-image data for Monkeypox diagnosis.

1.8 Document Structure

Chapter 2: A review of current studies on Monkeypox detection with deep learning is given in this chapter. CNN architectures, transfer learning strategies, dataset challenges, and explainability techniques including Grad-CAM are discussed.

Chapter 3: The method followed in this work is described in this chapter together with the datasets, preprocessing methods, and CNN-based models applied. It also addresses the evaluation criteria, data augmentation techniques, and transfer learning application.

Chapter 4 : This chapter offers a detailed about the development and implementation of the proposed system with CNN model, K-Fold cross-validation application, and Gradio-based interface integration.

Chapter 5: The results from the experiment are presented in this chapter together with evaluating model performance according to accuracy, sensitivity, specificity, and other evaluation standards. Included in addition are confusion matrices and graphical visualizations.

Chapter 6: The effects, advantages, and limitations of the findings are underlined in this chapter. It provides models performance and discusses its practical applicability in comparison with other research studies.

Chapter 7: Concluding and Future Research emphasizing its contributions to the identification of monkeypox, this chapter compiles the main conclusions of the research. It also identifies limitations and proposes directions for future research, including potential real world applications.

2

Literature Review

2.1 Advancements in Deep Learning for Medical Diagnostics

Machine learning (ML) and deep learning (DL) in particular have transformed medical diagnostics by allowing the automated examination of complex data. In the field of medical images, deep learning models have shown remarkable efficiency Figure 2.1 in identifying and categorizing diseases from image data. Convolutional neural networks (CNNs) a category of deep learning models have proven essential for their ability to comprehend visual data and extract hierarchical features (O'Shea & Nash, 2015).

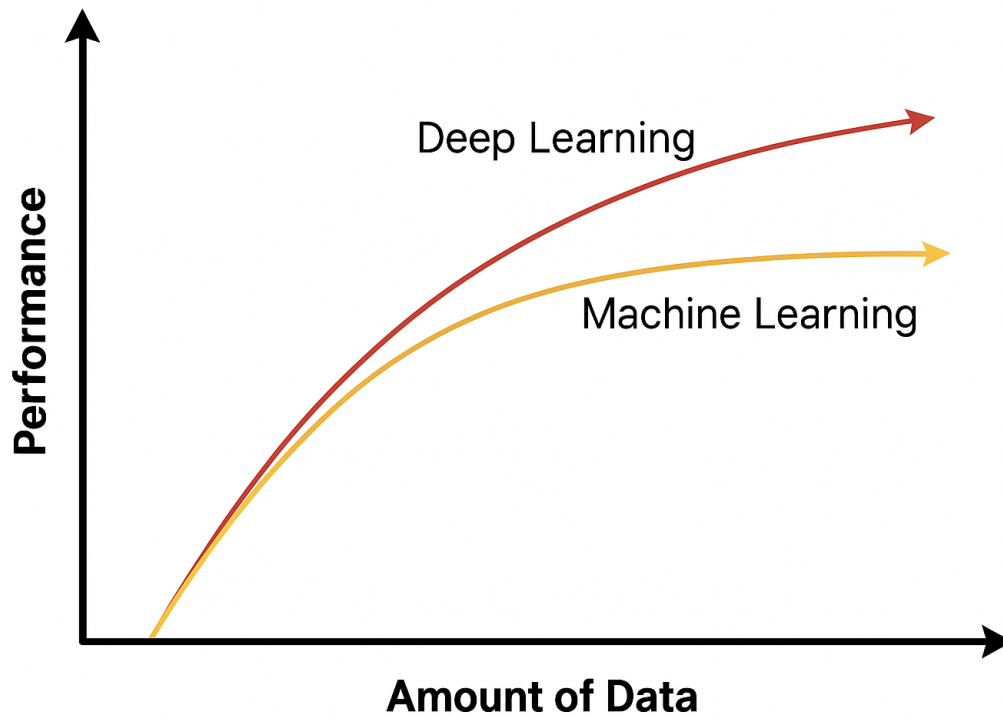


Figure 2.1: Deep learning vs Machine learning

2.1.1 Convolutional Neural Networks (CNNs) in Medical Imaging

Designed particularly for visual data analysis, CNNs are a type of artificial neural network architecture. Compared to traditional artificial neural networks, CNNs have greatly improved the processing, classification, and feature extraction capabilities of images. CNNs became popular for being able to handle the computational complexity connected with high dimensional data such as images. Unlike traditional artificial neural networks, CNNs use convolutional layers to reduce image processing parameters while maintaining the vital features needed for proper classification (O'Shea & Nash, 2015).

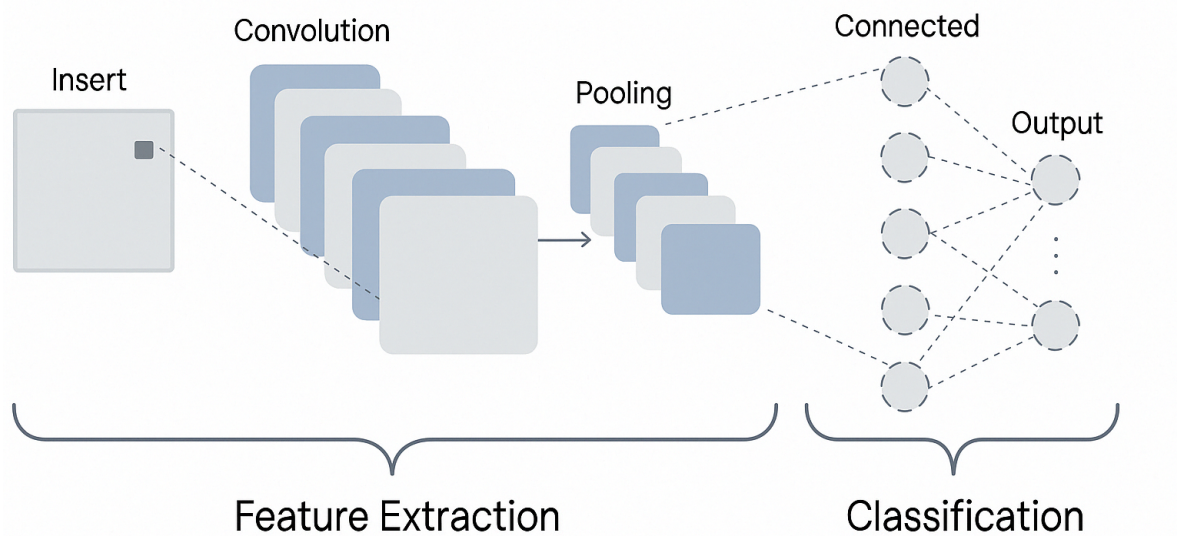


Figure 2.2: The structure of CNN

The human visual cortex inspires CNNs biologically therefore the structure of the network aims to mimic the hierarchical pattern of visual processing in the brain. They include a sequence of layers Figure 2.2 such as convolutional layers, pooling layers, and fully connected layers where each layer extracts deeper representations of the input image. CNNs are especially helpful for medical imaging, where changes in image characteristics can match important diagnostic information, such as differentiating between various skin lesions or identifying diseases such as monkeypox, which might visually resemble other skin conditions (O'Shea & Nash, 2015).

For image based tasks convolutional neural networks (CNNs) provide various benefits that greatly increase their efficacy. CNNs eliminate the need for human feature engineering by learning hierarchical spatial features directly from visual input so excelling in feature extraction. Using local receptive fields and weight sharing in convolutional layers helps them to drastically cut the parameter number hence enabling effective high dimensional images processing. CNNs also use weight sharing and pooling strategies that help to reduce overfitting and enhance generalization with strong design makes them especially efficient for image classification, object recognition, and segmentation also have the ability to manage noisy data ensures reliable results even under challenging.(O'Shea & Nash, 2015)

Nevertheless, CNN also has certain restrictions. Particularly with images with high resolutions they are computationally demanding since big filters and images consume memory consumption and processing time. Large labeled datasets are generally required for training CNNs these can be challenging to get and without enough data they are at risk of overfitting. CNNs for all their complexity are also impacted by problems such the vanishing gradient problem, which can impede appropriate weight updates and hinder learning. Furthermore, the large computing capacity required for training that

of high performance GPUs may restrict their use when faced with limited resources. Although CNNs offer unique advantages for image processing tasks and reliance on big datasets remain challenging overall (O'Shea & Nash, 2015).

2.1.2 Dataset Considerations in Medical Diagnostics

The availability of fully annotated datasets is one of the main difficulties in creating models for medical image analysis. With its hierarchical architecture and millions of annotated images, ImageNet has evolved into the foundation for transfer learning in medical diagnosis. Models are better able to generalize to medical datasets, which often lack the number and variety required to train strong models from scratch, using the characteristics learned in this huge dataset. (Deng et al., 2009) where concepts are organized into synsets connected by (IS-A) links. Aiming for 500–1000 photos per synset, it offers millions of high-resolution images spanning thousands of categories with careful human annotation producing great labeling accuracy. The scale and dataset variety help to train advanced models which often generalize effectively to other tasks. ImageNet is lacking segmentation annotations, stays incomplete (covering 10% of WordNet), and requires resources for training large models on it. Considering these difficulties it is still essential for improving object recognition and image classification research. (Deng et al., 2009)

2.1.3 Pre-trained CNN Models for Image-Based Diagnostics

Extensive architectures have been constructed and improved upon the fundamental knowledge of CNNs to tackle challenging image classification problems. Some well known CNN models and their applications to medical diagnostics are covered in the next subsections.

VGG Networks (Very Deep Convolutional Networks)

Mostly defined by their depth, which ranges from 11 to 19 weight layers, VGG networks also depend on the consistent use of 3×3 convolution filters. This design decision introduces more non-linearities but allows a stack of small filters to achieve the same receptive field as larger filters. Strong performance on the ILSVRC classification and localization tasks depends much on the depth of VGG networks, which exceeds past models, including those from ILSVRC-2012 and ILSVRC-2013. Their features translate well to different datasets, where handcrafted techniques generally fall short of representations, particularly the 4096-dimensional activations of the penultimate layer. Techniques including multi-scale training and evaluation, as well as the complementing application of dense and multi crop evaluation, help to further increase VGGs accuracy. Still this architecture has certain disadvantages even with multiple GPUs it is computationally expensive, requires significant training time, and the network depth causes high memory usage. The error rate seems to saturate at about 19 layers, and careful initialization is crucial pre-trained shallower networks often help increase convergence (Simonyan & Zisserman, 2014).

Building on this foundation, VGG networks introduced substantially deepening the architecture, marking a major step forward in CNN research to attain better accuracy, VGG employs a tiny convolutional filters (3×3) sequence and increases the depth of the networks to 16–19 layers. The consistent nature of VGG architectures makes them popular because of the simplicity with which they are built and expanded. Using VGG networks for medical diagnostics has shown that they can reliably identify small

details in images. However, they are not always the best choice for situations with limited resources or those requiring real time processing their relatively high computational requirements of approximately 15.9 to 19.6 billion FLOPS for VGG16 and VGG19 (Simonyan & Zisserman, 2014).

Residual Networks (ResNets)

Residual Networks ResNets were introduced by (He et al., 2016) to address a significant issue in deep learning: the performance degradation that occurs when additional layers are combined. ResNets implemented residual learning by employing shortcut connections and enabled the network to learn the residual function rather than directly fitting the underlying mapping. This framework simplifies the very deep networks training process, which can contain more than 100 layers, without the issue of vanishing gradients. The efficiency of the model is considerably enhanced by the three layer block of $(1 \times 1, 3 \times 3, 1 \times 1)$ convolutions in the bottleneck structure of ResNets (He et al., 2016) .

With the addition of residual learning, ResNets have become even more useful in medical diagnostics, as the ability to detect diseases early depends on distinguishing between healthy and diseased tissue and capacity to train deeper networks, ResNets can extract more complex and abstract features from medical images. With evidenced by their impressive performance in ImageNet competitions and their application to diseases such as monkeypox, where recognizing patterns in skin lesions can be challenging further demonstrate their capability in complex image analysis, ResNets, particularly those with pre activation, make it feasible to train networks with more than a thousand layers (He et al., 2016) .

By means of residual learning, where identity shortcut connections enable the network to learn residual functions rather than direct mappings, ResNets provide an effective solution to the degradation problem in very deep plain networks. While preserving simpler optimization and faster early stage convergence, this method enables architectures with depths reaching hundreds or even over a thousand layers. ResNets greatly enhance accuracy on tasks such as image classification, object detection, and localization, achieving remarkable performance, as evidenced by winning the ILSVRC 2015 competition with a 3.57% error rate. They also offer flexible designs (two-layer or three-layer residual blocks), allowing them to generalize successfully to a wide range of vision and non-vision problems. Very deep ResNets can have overfitting problems and their scale calls for computational resources, therefore careful initialization and learning rate changes are very important. Additionally, the use of bottleneck designs demands careful handling shortcut connections to prevent unnecessary complexity. Despite these constraints, ResNets set new benchmarks in accuracy and show amazing performance increases above related networks (He et al., 2016).

Densely Connected Networks (DenseNets)

By creating direct connections between all layers DenseNets solve the vanishing gradient problem and hence enhance strong feature propagation and information flow. Often using thin layers 12 filters and bottleneck layers to further lower complexity, DenseNets avoid relearning redundant features and achieve great parameter efficiency by concatenating feature maps from previous layers. Sometimes with less parameters than similar models, this architecture also provides an implicit deep supervision effect, helps normalize the model, and generally produces competitive or state of the art outcomes on benchmark datasets. DenseNets have certain shortcomings, they rely on transition layers for down sampling increasing complexity can still overfit in very deep configurations demand computational

resources and GPU memory for large-scale training, and suffer some implementation inefficiencies. By carefully setting hyperparameters and using growth rate, bottleneck layers, and compression, one can minimize these problems and expose the whole capability of DenseNets' parameter-efficient design. (Huang et al., 2017). DenseNets take advantage of feature reuse by concatenating feature maps that leading to architectures that are both compact and highly efficient. DenseNets use approximately a third of the parameters found in similar ResNets to achieve comparable performance levels with highlighting their impressive parameter efficiency. have been applied to tasks including skin lesion classification showing improves in both training stability and classification accuracy. Their dense interconnections provide an important advantage in detecting complex visual features related to skin diseases such as monkeypox and leading to more reliable diagnostic predictions than traditional CNNs. Furthermore, DenseNets have the ability to easily extend to hundreds of layers without facing optimization difficulties, which makes them particularly fitting for tasks that require in depth analysis of image features (Huang et al., 2017).

EfficientNet

Because of their compound scaling technique, EfficientNets are unique in that they achieve accuracy with significantly fewer parameters and FLOPs than previous convolutional architectures. This method ensures balanced expansion in all dimensions by consistently scaling the networks depth, width, and resolution according to a set of constants found using a small grid search. This leads to a family of models (EfficientNet-B0 through B7) that, although smaller in size, frequently outperform larger models in terms of inference time and accuracy. Their performance on a wide range of datasets with significantly fewer parameters than rival networks is further enhanced by their potent transfer-learning capabilities. EfficientNets significantly minimize redundant calculations by using an efficient baseline network EfficientNetB0 enhanced with mobile inverted bottleneck MBConv blocks and squeeze-and-excitation features. However, they also show certain limitations and dependencies: the accuracy gains depend on an efficient starting baseline, scaling coefficients must be adjusted via grid-search, and additional resolution or depth increases may be constrained by hardware memory ceilings. But still, EfficientNets architecture makes them a good choice among the convolutional neural networks available today for scenarios where efficiency and accuracy are essential.

EfficientNet has been implemented in medical diagnostics to diagnose diseases from images, providing high accuracy with relatively low computing expense. The architecture includes mobile inverted bottleneck blocks (MBConv), which are optimized for embedded or mobile applications. and the ability to balance accuracy, computational efficiency, and scalability makes it suitable for contexts with limited computational resources, such as developing regions or mobile diagnostic tools that help with early detection of diseases such as monkeypox (Tan & Le, 2019).

Inception Networks

By combining factorized convolutions, dimensionality reduction, and parallel modules, Inception networks reduce parameters and FLOPs without noticeably sacrificing representational power, making them effective in computationally limited settings The Inception or GoogLeNet it is developed by Google and introduced by (Szegedy et al., 2016), represents an important development in CNN architecture. Inception was developed to maximize the networks efficiency through its use of adjusted convolutions and dimensionality reductions that enable it to perform well even with computational and

memory limitations. Inception modules use different filter sizes to capture various features in parallel helps to capture both fine and coarse information simultaneously (Inceptionv3) improves the architecture by factorizing larger convolutions into smaller and more efficient processes, such as replacing a (5x5) convolution with two (3x3) convolutions and by exploiting asymmetric convolutions such as (1x7) followed by (7x1). Convergence and generalization are improved by techniques including label smoothing, grid size reduction, and auxiliary classifiers. However, scaling or adapting Inception-style models can be challenging modifications run the danger of undoing their computational benefits, and the original design concepts of the architecture may not be evident. Because doubling filter banks can double the number of parameters and computing cost, scaling becomes problematic. Furthermore, even while performance is reasonably resistant to changes in hyperparameters, overfitting is still a possibility, and careful tuning is necessary to extract the best accuracy. The Inception architectures ability to balance depth, width, and efficiency making it particularly suitable for medical applications where computational resources are limited yet accuracy is essential. and It has been applied to big data scenarios and mobile vision settings, showing its flexibility and practical application potential in medical imaging . (Szegedy et al., 2016),

2.2 Transfer Learning in Medical Imaging

Recent developments in Machine Learning (ML) and Deep Learning (DL) have demonstrated encouraging success in image analysis illness diagnosis. the transfer learning a branch of deep learning is being used more and more for identifying diseases as it handles intricate picture patterns with minimal datasets so effectively. well annotated datasets availability is one of the main challenges in creating models for medical image analysis. With its organized structure and millions of annotated images ImageNet has evolved into the pillar for transfer learning in medical diagnosis using features learned on this large dataset helps models generalize to medical datasets, which often lack the quantity and diversity required for training effective models from scratch. Particularly in cases with limited datasets, which is usually the case in medical research, transfer learning has become a main tool to use pre-trained CNN models. The fundamental concept is to fine tune a model on a smaller medical dataset using a pre-trained model on a large dataset like ImageNet. The approach benefits from the feature extraction capabilities learnt from millions of different photos and saves major computing resources and training time. (Deng et al., 2009),

2.2.1 Existing Approaches in Disease Diagnosis Using Image-Based Methods

1. **COVID-19 detection from X-ray images:** To identify COVID-19 in X-ray images, ,compared models like NASNetlarge and Inception-ResNet-V2 using transfer learning. It is essential to create automated diagnostic tools for viral disease, where early detection is critical, and these models showed great accuracy even with limited data availability, and the automation enables faster diagnosis for timely treatment. However, these models encounter difficulties because COVID-19 X-ray datasets are limited, leading to overfitting and underutilization of deeper models. Despite its usefulness, data augmentation cannot fully replace diverse, large scale datasets. Given these constraints, deep learning is still a promising method for COVID19 identification. Additionally, it may be difficult for previously trained models to adjust to imaging differences across different groups or equipment. (Albahli & Albattah, 2020).

2. Lyme disease Detection: The Images of erythema migrans (EM) skin lesions can be used to diagnose Lyme disease with the help of Convolutional Neural Networks (CNNs) using transfer learning. Although lightweight models like (EfficientNetB0) balance performance and efficiency, and pre-trained models like (ResNet, DenseNet, and EfficientNet) achieve high accuracy even with limited datasets making them appropriate for mobile devices and while enabling rapid self-assessment,. By showing lesion locations that affect predictions, tools such as Grad-CAM increase transparency while enabling non-expert decision making. These models have limitations, by their limited and undiversified datasets, which can lead to overfitting and decrease generalization particularly for darker skin tones that are underrepresented. The deep model computational complexity such as ResNet50 restricts their application in environments with limited resources, and the lesion segmentation makes it harder to concentrate on the affected regions. CNNs are still a useful tool for early Lyme disease identification in spite of these obstacles while dataset accessibility and model deployment should be enhanced. (Hossain et al., 2022).

3. Detection of Alzheimers disease: Detecting Alzheimers disease using medical images such as MRI scans has shown great benefits from both CNNs and transfer learning. Using pre trained architectures like VGG16, VGG19, and MSCNet and automating feature extraction, these models produce great accuracy. For gray matter segmentation, MSCNet attained 98.85%, for instance; VGG based models excel in accuracy and recall for Alzheimers disease detection. Transfer learning is effective for medical image analysis since it lowers the demand for large amounts of processing resources and training data. These strategies have restrictions, though Real world applications may find CNNs great computational capability and strong hardware challenge problematic. Often the result of data privacy concerns, small, non diverse datasets can cause overfitting and lower generalizability. Moreover, CNNs black-box character makes it challenging for doctors to understand the acquired properties, therefore undermining confidence in the forecasts. Although performance is encouraging, more general clinical acceptance depends on changes in dataset availability and model clarity. (Salehi et al., 2023) .

2.2.2 Existing Approaches and Comparative Insights in Monkeypox Disease Diagnosis Imaging

One of the most popular methods for early diagnosis in monkeypox image detection is transfer learning allows to use Convolutional Neural Networks (CNNs) that have been trained on large datasets, which is useful since there are not many monkeypox-specific data provided. Using this approach the models may effectively detect monkeypox from other diseases of the skin through using learned features from large picture libraries.

Building on this concept,(Ahsan et al., 2022) expanded on this idea by testing monkeypox detection using models such as ResNet50, MobileNetV2 and VGG16. Using transfer learning, these models were able to identify monkeypox from other skin diseases with accuracies between 93% and 99%. The results of this study confirm that transfer learning works even when there is a lack of data . Rotation, shifting, and flipping are among the data augmentation methods used to diversify the dataset and solve the problems related to the small size. In particular cases, such as differentiating monkeypox from chickenpox, the implementation of majority voting improves classification accuracy even more. Furthermore, it offers thorough performance based on criteria including precision, recall, and F1-scores, therefore revealing the strengths and shortcomings of several models(Ahsan et al., 2022).

though as well. The limited generalization of the results in the small and imbalanced dataset ; the models struggle with subtle interclass differences, as observed in the rather poor performance in identifying monkeypox from measles. Although majority voting produces average results in challenging classifications, it improves some outcomes. Moreover, the study ignores explainability methods, which are absolutely important for medical uses to ensure model interpretability and build trust among medical experts. Furthermore impeding repeatability and independent validation of the results is the lack of publicly accessible databases. Dealing with these limitations would greatly improve the relevance and practical use of the study.(Ahsan et al., 2022)

Expanding the exploration of transfer learning techniques,(Ahsan et al., 2023) presented a Generalization and Regularization-based Transfer Learning Approach (GRA-TLA) for monkeypox diagnosis. After evaluating ten different CNN models, they found that ResNet-101 achieved multiclass classification accuracy rates up to 99% and that Xception was the best at binary classification with accuracies ranging from 77% to 88%. More importantly for medical applications, the implementation of Local Interpretable Model-Agnostic Explanations (LIME) improved the interpretability of model predictions by drawing attention to specific features indicative of monkeypox.

Similarly,This (Jaradat et al., 2023) emphasizes major developments and difficulties in automated deep learning and transfer learning based monkeypox detection. The performance of the MobileNetV2 model, which attained high accuracy (98.16%), precision (0.99), recall (0.96), and F1-score (0.98), so demonstrating its efficacy in classifying monkeypox from skin lesion images. Along with transfer learning to improve accuracy despite a small dataset, the study also used pretrained models like VGG16, VGG19, ResNet50, and EfficientNetB3. Robust evaluation criteria and advanced data augmentation techniques (rotation, flipping, zooming) further helped the model's generalization and alignment with the key need for fast diagnosis. Still, the limited collection of 117 images in the study reduces its generalizability to many real world scenarios. The reliability of internally gathered data without outside validation begs questions regarding its general applicability. The dataset size runs the risk of overfitting even with augmentation, so limiting reliability in clinical settings. Moreover, the research skipped alternative machine learning methods; the validation accuracy of MobileNetV2 declined somewhat (94%) on outside datasets, suggesting opportunity for development. Although the studies show a good overall strategy, they underline the requirement of larger datasets as well as external validation to ensure efficient implementation.(Jaradat et al., 2023)

In line with this perspective (Abbaraju et al., 2023) explored the Particularly noting the diagnostic difficulty resulting from symptom similarities with other diseases and limited knowledge of rare diseases, the research successfully solves the important challenge of early monkeypox identification, necessary for prompt treatment and disease management. The research shows the potential of such models for medical applications by using advanced methods including deep learning (DL) and machine learning (ML), especially CNNs, which have shown successful in other settings. From Kaggle, the Monkeypox Skin Image Dataset was expanded using data augmentation methods like rotation, downsizing, flipping, and shearing, therefore addressing the data scarcity. Using pre-trained networks—such as those trained on the IMAGENET dataset transfer learning allowed pre-trained networks to adjust to monkeypox classification tasks. With models like VGG19 and ResNet50, the study gave a thorough assessment of models utilizing metrics including accuracy, precision, recall, and F1-score, thereby attaining a high recognition accuracy of 92%. Together with Support Vector Machine (SVM), a comparison of many models including VGG16, VGG19, ResNet50, and EfficientNet highlighted the most

successful strategies. The study has few limitations too, including the limited starting dataset (228 images), which might limit generalizability, and the binary classification focus, which keeps difference among different skin diseases out of mind. Although technologies like Grad-CAM and LIME help to improve explainability, the inherent opacity of deep learning models remains a problem along with possible misclassification risks from limited datasets. Moreover, ethical issues regarding the use of patient data still exist and highlight the need of cautious application and validation in practical situations.(Abbaraju et al., 2023)

2.3 Hybrid Feature and Ensemble Techniques in Monkeypox Detection

2.3.1 Hybrid Feature

With the objective to improve the accuracy of feature extraction and classification for early monkeypox detection, the (Maqsood et al., 2024) MOX NET uses a transfer learning method and utilizing pre-trained deep learning models. Robust feature vectors were extracted from the monkeypox image dataset applying six pretrained models: ViT, Swin Transformers, ResNet-50, ResNet-101, Efficient-NetV2, and ConvNeXt-V2. The models were fine tuned to recognize between normal skin, monkeypox, chickenpox, and measles following being trained on big datasets like ImageNet. To strengthen the models ability to deal with image variability that is inherent to medical imaging, the architecture was fine tuned by adding additional layers that were suitable for the classification task and freezing existing layers. After all six pretrained models have finished feature extraction, a convolutional sparse image decomposition (CSID) fusion technique was used to merge the deep feature vectors, keeping the best features from each model. To further optimize the resulting feature set to improve computational performance, an Entropy-Controlled Firefly (ECF) feature selection method was employed to reduce redundant features. The images were categorized into four groups: Monkeypox, chickenpox, measles, and normal using a multiclass support vector machine (M-SVM) that was fed the features that were chosen.(Maqsood et al., 2024)

Similarly,(Bala et al., 2023) employ transfer learning through a modified DenseNet201 model. The article presents the development of MonkeyNet, a model designed to improve the detection of Monkeypox skin images. DenseNet201 was selected as the base model because of its efficient feature reuse and ability to address the vanishing gradient issue making it ideal for complex medical image processing and trained on large datasets such as ImageNet, DenseNet201 provided MonkeyNet with a solid foundation for extracting essential features from skin lesion images. The authors modified DenseNet201 for Monkeypox detection by including new fully connected layers intended to identify four categories: Monkeypox, chickenpox, measles, and normal skin. The adjustments included batch normalization to enhance training stability and dropout layers to reduce overfitting. The model was fine-tuned to learn from the recently developed “Monkeypox Skin Images Dataset (MSID) containing curated and augmented images. MonkeyNet effectively collected features, including textures and lesion patterns, which were crucial to identifying similar skin diseases. It achieved 93.19% accuracy on the original dataset and 98.91% on the augmented dataset, showing its strong capability in early monkeypox detection. (Bala et al., 2023)

performed a comprehensive evaluation of different pretrained CNN models, Their findings demonstrated that the VGG16 and VGG19 models showed competitive performance, achieving up to 97%

accuracy in binary classification tasks including the separating between Monkeypox and chickenpox. ResNet50 had major accuracy in identifying Monkeypox from non Monkeypox diseases, comparable to architectures such as InceptionV3 and DenseNet. In particular, DenseNet showed great results by achieving an F1-score of 99% to distinguish Monkeypox from Chickenpox. This proves that DenseNet could be particularly effective in feature propagation and reuse essential for identifying diseases with visually similar symptoms (Saha et al., 2023). Models like (Xception and MobileNetV2) show performance in scenarios with limited computational resources making them suitable for mobile diagnostic applications. On the other hand, LeNet an early CNN architecture showed considerable performance in detecting Monkeypox from other diseases.(Saha et al., 2023) Furthermore, a majority voting approach it been used where predictions from multiple models are aggregated achieving a 97% accuracy rate in differentiating between Monkeypox and Chickenpox. This combining model highlights the benefits to improving overall accuracy and adaptability in difficult scenarios involving limited or imbalanced datasets (Saha et al., 2023).

2.3.2 Ensemble Learning

Ensemble techniques are designed to improve model performance by integrating the predictions of multiple models, while hybrid feature fusion focuses on the combination of features and methods that substantially improve CNN-based models for the early detection of Monkeypox accuracy, robustness, and generalization. Bagging is an ensemble method that is effective it entails the training of multiple models on various bootstrapped samples of the dataset this methodology mitigates overfitting and documents a variety of patterns in the data so Bagging reduces variance and establishes a more dependable model for distinguishing between Monkeypox and non Monkeypox cutaneous lesions by integrating predictions through majority voting (Ganaie et al., 2022). (Pramanik et al., 2023) took this idea even further and suggested an ensemble method that would combine the results of the InceptionV3, DenseNet169 and Xception models that were trained on a data set of images of monkeypox skin. Combining these models allowed the ensemble to outperform individual base learners in terms of generalization capacity by averaging 93.39%. Beta function-based normalizing helped to further improve probability aggregation, thus producing more accurate forecasts (Pramanik et al., 2023) .

(Haque et al., 2023) similarly investigated the efficiency of an ensemble model applying majority voting for monkeypox detection. In data sets including monkeypox, measles, chickenpox and normal skin photos, the team aggregated the predictions of five models DenseNet121, ResNet50, InceptionV3, EfficientNetV2B3 and ResNet152V On the MSID dataset, the ensemble model attained an accuracy of 98.7% by performing this high level of performance shows how very successful the majority voting strategy was in improving the detection accuracy.

In particular in the imbalanced datasets observed in their comparative investigation, ensemble techniques often exceeded individual models. They used stratified cross-valuation to guarantee that every dataset fold was representative, hence reducing the effect of data imbalance. With regard to general classification measures, the ensemble model outperformed individual models, offering convincing proof of the application of ensemble methods in medical image analysis (Haque et al., 2023).

Putting emphasis on the varied data value, trained and evaluated their ensemble model using two datasets, the (Monkeypox Skin Images Dataset (MSID) and the Monkeypox-dataset-2022 (MD-2022)).

In addition, stratified cross valuation guaranteed strength over different data subsets. In the MSID data set, the final model achieved an accuracy of 98. 7% and in the MD 2022 dataset 89.4% This highlighted the need to validate the performance of the model in several contexts to ensure generalizability. (Haque et al., 2023)

From a practical standpoint implementing these ensemble approaches for early monkeypox detection means augmentation of data with enough diversity. Although decision fusion methods efficiently combine their outputs and techniques like stacking or boosting help to further improve the ensemble models. These ensemble methods improve data imbalance, generalization management and provide more resilience they ultimately result in more precise and consistent identification of monkeypox is crucial for immediate action and disease control (Ganaie et al., 2022).

2.4 Comparative Analysis of Studies for Monkeypox Disease Diagnosis Using CNN Approaches

Reference	Contributions	Algorithms Used	Performance Metrics	Explainability	Generalization	Classification Type
(Ahsan et al., 2022)	Proposed modified CNN architectures for Monkeypox detection	VGG16-VGG19-InceptionResNetV2-ResNet50-ResNet101-MobileNetV2	MobileNetV2 (Acc:99%) InceptionResNetV2 (Acc:100%)	LIME	Yes	Binary
(Ahsan et al., 2023)	(GRA-TLA) for binary and multiclass classification of monkeypox using 10 CNN models.	VGG16-ResNet50-ResNet101-Xception-EfficientNetB0-EfficientNetB7-NasNetLarge-EfficientNetV2M-ResNet152V2-EfficientNetV2L	Binary:Xception(acc:88%) - Multiclass: ResNet101 (acc:99%)	Grad-CAM++,LIME	No	Binary
(Alhasson et al., 2023)	Mobile based deep learning model	MobileNetV2	Acc: 99%	No	Yes	Binary
(Altun et al., 2023)	Customized CNN with transfer learning and hyperparameter optimization.	MobileNetV3s-EfficientNetV2-ResNet50-VGG19-DenseNet121-Xception	MobileNetV3s (Acc: 96%)	No	Yes	Binary
(Sorayaie Azar et al., 2023)	Developed seven DNNs for Monkeypox detection	DenseNet201-InceptionResNetV2-InceptionV3-ResNet152V2-VGG16-VGG19-Xception	Binary (Acc:97.63%-)Multiclass (Acc:95.18%)	LIME, Grad-CAM	Yes	Multiclass-Binary
(Bala et al., 2023)	Proposed a novel CNN model (MonkeyNet) for Monkeypox	DenseNet201 (MonkeyNet architecture)	Original dataset: 91.91% Augmented dataset: 98.91%	Grad-CAM	Yes	Multiclass
(Haque et al., 2023)	Ensemble model with majority voting for monkeypox detection Compared five DL models	DenseNet121-ResNet50-ResNet152v2-InceptionV3-EfficientNetV2B3	MSID: 98.7% (Ensemble), MD2022: 89.4%(Ensemble),	No	Yes	Multiclass
(Jaradat et al., 2023)	Evaluated five pretrained DL models for binary classification of monkeypox.	VGG19-VGG16-ResNet50-MobileNetV2-EfficientNetB3	MobileNetV2 (Acc:98.16%)	No	Yes	Binary
(Maqsood et al., 2024)	MOX-NET combining six DL models (hybrid feature fusion)	ViT, Swin Transformer, ResNet50-ResNet101-EfficientNetV2-ConvNeXtV2	Acc: 98.64%	Grad-CAM	Yes	Multiclass
(Nayak et al., 2023)	Evaluated five DL models for Monkeypox detection	Places365-GoogLeNet-SqueezeNet-AlexNet-ResNet18	ResNet-18 (Acc:99.49%)	LIME, Grad-CAM	Yes	Binary
(Pramanik et al., 2023)	Proposed a Beta function-based ensemble learning framework for Monkeypox detection using three CNN models	InceptionV3-Xception-DenseNet169	Average acc:93.39% (5-fold cross-validation)	Grad-CAM	Yes	Binary
(Saha et al., 2023)	Compared CNN-based pre-trained models for monkeypox detection Demonstrated majority voting approach	VGG16-VGG19-ResNet50-InceptionV3-DenseNet-Xception-MobileNetV2-AlexNet-LeNet	- 97% (Monkeypox vs. Chickenpox) -96% (Monkeypox vs.Normal) -79% (Monkeypox vs.Measles) -80% (Monkeypox vs.All)	No	Yes	Binary
(Abbaraju et al., 2023)	Developed a prediction model using various DL algorithms	SVM-CNN-VGG(16-19)-ResNet(50-101)-EfficientNet(B0-B1-B2)	VGG19 and ResNet50 achieved 92% acc	LIME , Grad-CAM	Yes	Binary

Table 2.1: Comparative Analysis of Studies on Monkeypox Detection

2.5 Explainability and Interpretability in deep Learning Models : The Role of Grad-CAM

Deep learning models have shown remarkable performance in several fields, including healthcare, where they are used mainly in disease diagnosis. However, their black-box, especially in important uses like Monkeypox diagnosis, makes their decision-making process difficult to grasp. Gradient-weighted Class Activation Mapping Grad-CAM has become a fundamental instrument for improving model transparency by visually highlighting the areas of input data most likely to affect the predictions of a model. In order to investigate the function of Grad-CAM in enhancing the interpretability of deep learning models, especially in Monkeypox detection using skin lesion images, this work examines and synthesizes ideas from several investigations.

2.5.1 The Role of Explainability in Deep Learning

To attain great accuracy deep learning models depend on intricate layers and parameters. Still, this complexity sometimes hides how models come to their choices. Grad-CAM and other explainability techniques overcome this restriction by creating heatmaps localizing the areas of an input image most pertinent to a forecast. In medical uses, where accurate and interpretable forecasts can foster trust and help decision making, this capacity is especially important in identifying Monkeypox lesions from other skin conditions.(Altun et al., 2023; Nayak et al., 2023)

2.5.2 Grad-CAM: A Tool for Explainability

Leveraging gradient information from the last convolutional layers of a CNN, Grad-CAM creates visual explanations. These visualizations heatmaps show the important locations in an input image that affected the prediction of the model, therefore providing class-specific visuals.

Grad-CAM has several benefits. It lets doctors grasp the model's decision-making process by means of simple and understandable visuals that highlight the areas in an image accountable for a prediction (Altun et al., 2023; Sorayaie Azar et al., 2023). Without architectural changes, it is also model agnostic and compatible with other architectures like DenseNet 201, ResNet50, MobileNetV3, and VGG16 (Altun et al., 2023; Bala et al., 2023). Grad-CAM is flexible in uses including identifying between Monkeypox, chickenpox, measles, and normal skin disorders (Bala et al., 2023; Nayak et al., 2023). It handles both binary and multiclassification tasks. Furthermore, it enhances the whole knowledge of model predictions by complementing other interpretability methods such as Local Interpretable Model-Agnostic Explanations (LIME), thereby improving the understanding of model predictions(Altun et al., 2023).

At the same times the Grad-CAM has limitations, The heatmaps it creates are sometimes coarse and lack of fine-grained details, which would restrict their applicability for exact clinical diagnosis(Nayak et al., 2023; Sorayaie Azar et al., 2023). Particularly when gradients are sparse or unstable, its reliance on gradient information might cause noise or errors in the visualizations (Bala et al., 2023). Furthermore adding variability and subjectivity is Grad-CAM explanations, which are intrinsically qualitative and demand expert interpretation (Altun et al., 2023). Finally, Grad-CAM is limited to convolutional layers and does not consider features learnt in fully connected or transformer-based architectures, therefore restricting its application in some circumstances(Selvaraju et al., 2016).

2.5.3 Applications of Grad-CAM in Monkeypox Detection

Grad-CAM has been extensively applied to improve interpretability and confirm model predictions in the detection of Monkeypox. Among the important uses is the location of the affected areas. Grad-CAM uses heatmaps to show skin lesion areas most pertinent to forecasts, thereby helping doctors to evaluate the emphasis of a models attention (Altun et al., 2023; Sorayaie Azar et al., 2023). Models such as ResNet50 and DenseNet 201 used Grad-CAM, for instance, to visually verify that their predictions were derived from medically relevant data.

Grad-CAM is also utilized for the validation and comparison of several CNN designs. Higher performance models such DenseNet 201 and MobileNet-V3s concentrated more precisely on lesion areas than others including VGG16 and Xception (Bala et al., 2023). Moreover, Grad-CAM has shown good performance in actual implementation, especially in environments with restricted resources. MobileNetV3 transformed to TensorFlow Lite (TFLite) was included into mobile diagnostic apps, therefore enabling real time, easily accessible Monkeypox detection (Altun et al., 2023).

Furthermore, Grad-CAM offers explainability for transfer learning models, hence complementing them. Transfer learning models like MobileNetV3s and EfficientNetV2s, for example, used Grad-CAM heatmaps to test their conclusions and attained F1-scores above 97% thereby assuring that predictions were based on pertinent features (Altun et al., 2023) (Nayak et al., 2023).

at Final Thought the Grad-CAM offers visual explanations that improve the interpretability of deep learning models. Its use in the diagnosis of Monkeypox has shown great importance since it helps to validate model predictions, localize disease specific traits, and raise diagnostic confidence. Although Grad-CAM has drawbacks including coarse heatmaps and reliance on gradients, its interoperability with many architectures and connection with other tools like LIME emphasize its potential to progress. Researchers might increase the value of deep learning models in important fields like healthcare by overcoming these constraints and aggregating Grad-CAM with complementary approaches.

3

Methodology

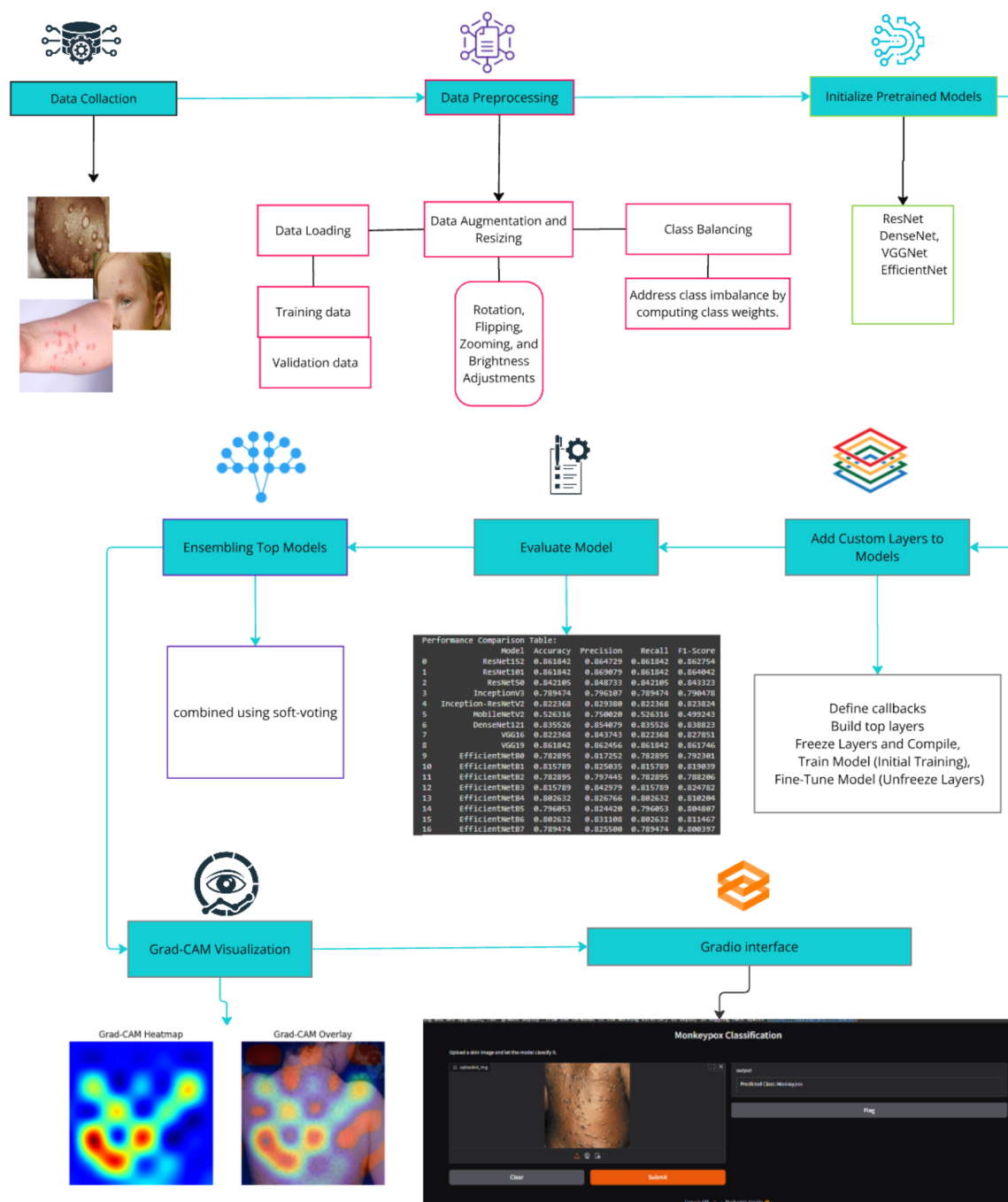


Figure 3.1: Flow diagram used in this research to classify Monkeypox

3.1 Data Collection

The methodology begins with thorough **Data collecting** as shown in Figure 3.1, the approach include a several skin disease images (Normal, Monkeypox, Chickenpox, and Measles) collected from Kaggle dataset (Bala, 2022). with class sizes of 293 (Normal), 279 (Monkeypox), 107 (Chickenpox), and 91 (Measles), as shown in Figure 3.2. :

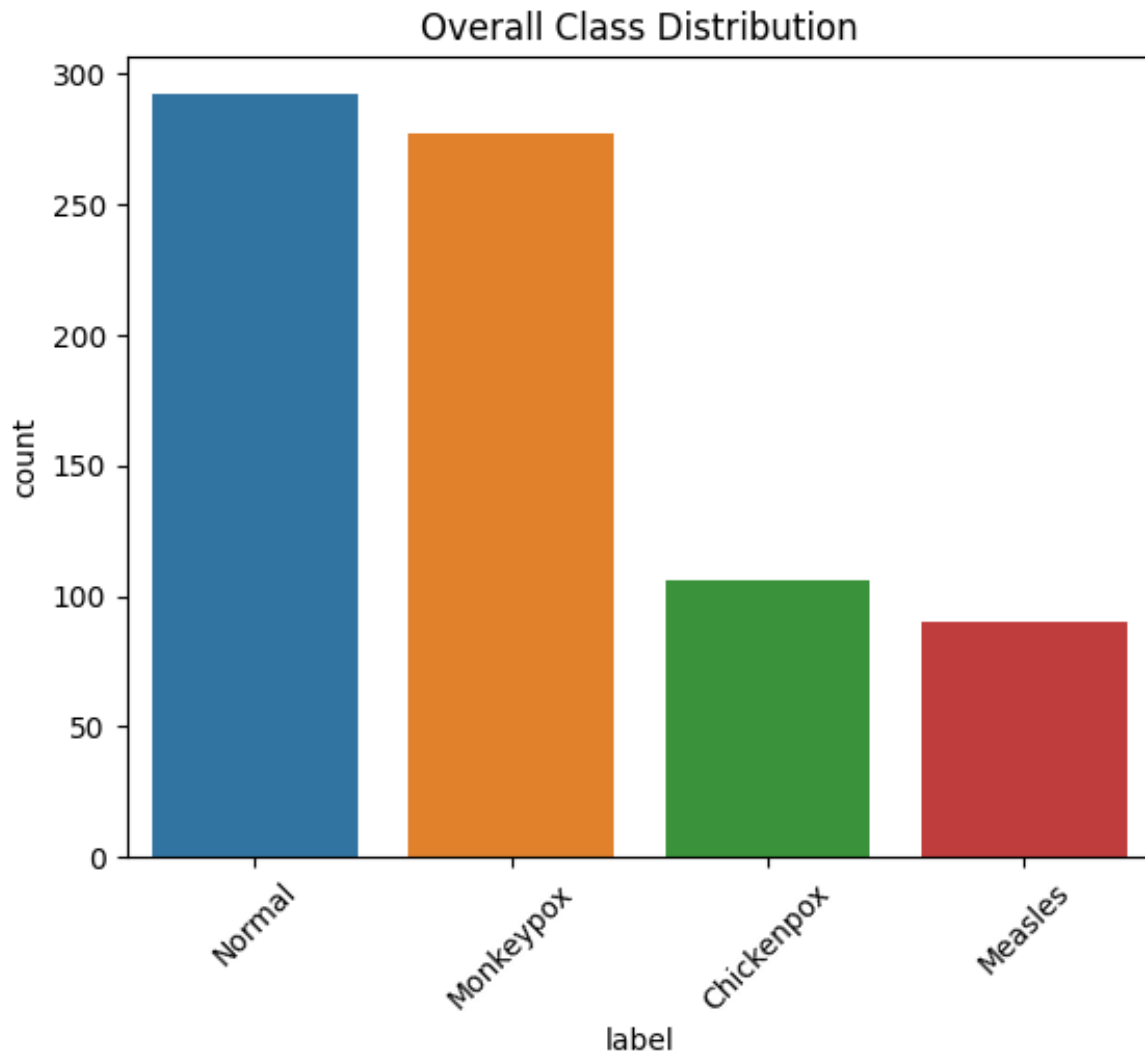


Figure 3.2: Dataset Class Distribution

The Normal and Monkeypox categories each account for approximately 36–38% of the data, whereas Chickenpox and Measles each include roughly 12–14%. With the smaller classes not being very small a noticeable imbalance persists that may impact the models ability to learn and generalize.

3.2 Data Preprocessing

Before starting the **Data Preprocessing** these images were organized into matching class-label directories. This includes cleaning that is, deleting duplicates or poor quality images and normalizing

them to a standard size typically 224×224 for most CNN designs. As can be observed from Figure 3.1 the dataset was split into 80% for training and 20% for validation, therefore allocating most of the data to model training and leaving a smaller fraction for hyperparameter tuning and performance. Because the model is periodically evaluated on unprocessed data during the training part, this configuration helps avoid overfitting. This part split is a useful starting point, particularly if an external test set or cross validation will be added later for more thorough performance evaluation some processes also include a final test set separate from both training and validation. **Data augmentation** is applied to the training set only typical, transformations include random rotations, flips, zooming, shifts, and adjustments in brightness or contrast, each applied probabilistically to increase the effective training sample size and so reduce overfitting and improve model generalization. Class weighting is calculated in case the dataset is class-imbalanced, therefore ensuring minority classes such as Monkeypox should it have fewer samples and more relevance in the training goal.

3.3 Model Selection and Architecture

Following that **Model Selection and Initialization**, multiple pretrained Convolutional Neural Network architectures are chosen, typically including **ResNet**, **DenseNet**, **VGGNet**, **MobileNet**, **Xception**, **Inception**, and **EfficientNet**. These models, initially trained on the extensive ImageNet dataset, are loaded without their top (fully connected) layers. This step generates a basic model of convolutional and pooling layers that collect extensive, generalized visual information suitable for transfer learning.

Additional layers are added to fit the particular skin lesion categorization task from the basic models. Typically utilized for reducing spatial dimensions, a **GlobalAveragePooling2D** layer comes first followed by a fully connected Dense layer (**256 units, ReLU activation**) and a **Dropout layer (0.2)** to help to lower overfitting. Using **softmax** activation and a neuron count matching the classes number, the last Dense layer allows multi class probabilities to be predicted.

3.4 Two-Phase Training Process:

Training proceeds in two distinct phases to optimize the transfer learning benefit:

3.4.1 Phase 1: Top-Layer Training

The ImageNet learned weights of the base model remained **frozen**. Trained with faster learning rate that is, **1e-4** only the recently added layers. This stage tunes the top layers to identify patterns unique to Monkeypox and associated lesions.

3.4.2 Phase 2: (Fine-Tuning)

After initial training, the base model or select deeper layers is **unfrozen** after the custom layers are partly matched with the medical data. The lower level convolutional filters are softly adjusted using a lowered learning rate, like **1e-5**. Since more of the network parameters fit the new domain, this fine tuning usually produces additional increases in accuracy and generalization.

3.5 Training Management and Evaluation

Throughout these phases, callbacks help manage the training workflow **EarlyStopping** stops training ...if the validation loss fails to improving, hence preventing overfitting. Every time a performance improvement on the validation set is seen, **ModelCheckpoint** saves the best weights for the model. If validation loss plateaus, **ReduceLROnPlateau** lowers the learning rate allowing the model to escape local minima.

Assessed performance then comes from the validation or test set. Generated is a classification report with accuracy overall, F1-score, precision, and recall. Raw and adjusted confusion matrices both expose per-class performance and possible regions of misclassification. Tracking accuracy and loss for both training and validation periods, Learning curves show whether the model underfits or overfits.

3.6 Model Interpretability with Grad-CAM

Interpreting a medical application is crucial. The salient areas of every image that affected the forecast of the model are therefore highlighted using **Grad-CAM visualization**, as show in Figure 3.1 . This overlay of heatmaps on to the image helps verify whether the models attention corresponds with clinically relevant features.

3.7 Model Ensembling

Ensembling Top Models often provides better prediction power since individual models capture unique characteristics. Based on their F1-scores or another preferred metric, the top performers load and their results are aggregated with **soft-voting**. Often resulting in increased robustness and overall accuracy, by averaging the output probabilities of every model for a given image, they can take advantage of the complementing strengths of the constituent designs.

3.8 Deployment via Gradio Interface

Finally, following the model or ensemble is chosen, a **Gradio interface** (Abid et al., 2019) can be developed. Clinicians or researchers can upload new images and instantly get categorization predictions using this easy web-based tool. Supporting explainability such Grad-CAM heatmaps, one can provide real-time evaluation and validation of decisions In clinical or research environments.

In summary, the methodology was able to create an efficient, and user-friendly framework for Monkeypox image classification by including all these components of data handling, augmentation, structured two-phase training, explainable visualization, ensembling for improved accuracy, and a practical Gradio-based interface.

4

Implementation

4.1 Introduction to the Implementation

Medical image classification requires careful data management and model requirements handling. From an analytical perspective, the focus is on data integrity confirming that images reflect targeted classes, finding duplication, and applying modifications producing noticeable results. , an implementation driven perspective stresses how leverage deep learning libraries, follow design principles, implement automation cross-valuation, ensure interpretability, and finally apply the system for useful purposes.

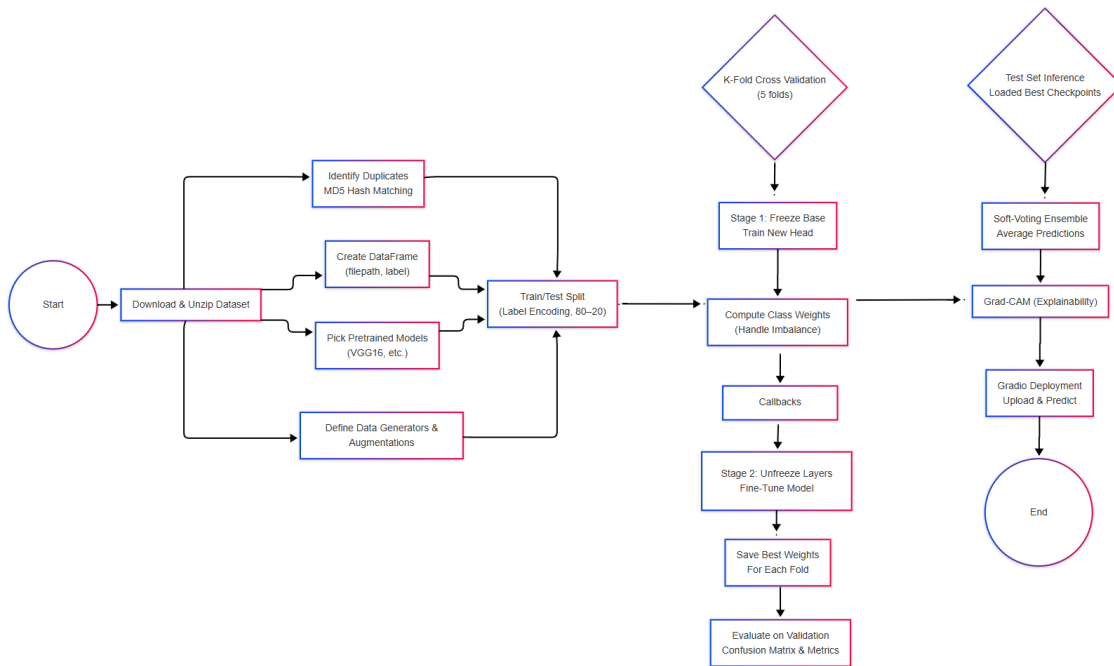


Figure 4.1: Workflow diagram

As show on Figure 4.1 these three stages mix in this chapter: Start by outlining how and why data was gathered and cleaned and then turn to the deep learning specifics pipeline, including neural designs, training schedules, and performance. At last, discuss explainable and detail Gradio integration, a user friendly technique that made classifier accessible to a larger, maybe non technical audience.

4.1.1 Dataset Overview

Accurate identifying Monkeypox required collecting images of relevant skin lesions along with other diseases (like chickenpox or measles) and normal, healthy skin samples. In the medical industry, a dataset this size is naturally difficult to gather for ethical, logistical, and legal considerations. Luckily, a publicly available Monkeypox Skin Image Dataset on Kaggle (Bala, 2022) supplied a well chosen collection of tagged images. The Kaggle repository reduces data collecting effort by aggregating contributions from several public-domain medical image sources.

In addition, the dataset includes 770 images and pre approved labels: (Monkeypox, Measles, Chickenpox, and Normal). This multi class approach provides the basis for a more flexible classifier that can differentiate between related or maybe confusing circumstances in addition to detecting monkeypox in isolation.

4.1.2 Data Download Procedure

Originally, data was obtained using the Kaggle API (Bala, 2022) a command line interface that lets datasets be directly downloaded into cloud-based system. Following a successful download, the compressed zip file was unzipped showing a directory structure in which every class label matched a subdirectory.


```
Monkeypox Skin Image Dataset/  
  Monkeypox/  
    (image files)  
  Measles/  
    (image files)  
  Chickenpox/  
    (image files)  
  Normal/  
    (image files)
```

Since folder names matched the classes, this structure allowed for a simple connection between every image and its presumed label. To match the predicted amount from Kaggle, the total file count was checked and images from each subdirectory were randomly selected to verify that files loaded correctly and looked consistent with their stated labels.

4.2 Data Cleaning and Processing

4.2.1 Duplicate Removal

Duplicate pictures are an risk in medical imaging datasets since they may cause a classifier to be overestimated. A model can efficiently cheat by memorizing those repeated patterns when duplicates show up in both training and validation sets, therefore inflating its validation accuracy. With set up an MD5 hash based detection method to reduce this risk. For every file path opened the image in binary mode, calculated its MD5 hash, and kept the output in a dictionary keyed by hash values.



Figure 4.2: Detection Duplicate In Dataset

The image was flagged as show 4.2 a duplicate noting both the new and original path if a newly computed hash already existed in the dictionary and showing the suspected duplicates side by side

served as a last check. In practically all hash collisions, the images were exactly copies. Although additional preparation time, this technique was deemed essential. In a medical environment even a few duplicated samples may skew findings. Once duplicates were found and verified they were removed in order to keep the dataset's authenticity and reliability.

4.2.2 Label Encoding and DataFrame Creation

Following data validity confirmation, and before passing labels into neural networks, one typically transforms them into text labels into numerical indices. Using a Label Encoder from scikit-learn, the four category names (Monkeypox, Chickenpox, Measles and Normal) were mapped to integers (0, 1, 2, 3). Simplifying subsequent data splits and transformations that stored all file paths and related numerical labels in a Pandas DataFrame that offers a range functionalities for grouping, filtering, and sorting, therefore enabling versatile dataset management.

4.2.3 Train–Validation–Test Split

Furthermore, the dataset is split into training, validation, and test sets. Specifically, a stratified split at 20% ensures that the same class distribution is preserved across the folds, and the test set was isolated from all model experiments until the final evaluation preserving its role as an unbiased gauge of performance. The code snippet below follows this approach:

```
df_train_val, df_test = train_test_split(
    df_all,
    test_size=0.2,
    stratify=df_all['label_idx'],
    random_state=42)
```

In medical datasets, stratification is essential since some diseases might have fewer images ensuring a representative proportion in every subset, thus avoiding misleading results and supporting improved generalization on data with reasonable class ratios. This approach ensures that the final performance measures that are given closely reflect the model's actual capacity to categorize unseen data. Additionally, it prevents by mistake optimizing test set hyperparameters, which would compromise result validity.

4.2.4 Addressing Limited Data with Augmentation:

Data lack is usually a major problem in medical imaging. In general, high-quality labeled images are costly and require professional annotation, specialized equipment, or time. Augmenting existing images can help reduce this issue. The Keras (ImageDataGenerator) class is set up to carry out random transformations in this project, As showing in table 4.1 including flips, zooms, rotations, and width/height shifts. By artificially increasing the dataset, these changes train the model to deal with changes in positioning, scale, and angle. Applying these changes increases the neural networks effectively sees a slightly different version of each image at every epoch, thereby reducing overfitting and robustness by exposing it to a wider range of inputs. Image augmentation is very simple. It's important to use caution when applying too many changes so that the images unrealistically differ from real world clinical situations.

Parameter	Value	Description
Rotation Range	30°	Maximum random rotation
Zoom Range	0.3	Random zoom in/out factor
Width Shift Range	0.2	Fraction of total width for horizontal shifting
Height Shift Range	0.2	Fraction of total height for vertical shifting
Horizontal Flip	True	Randomly flip images horizontally
Vertical Flip	True	Randomly flip images vertically
Preprocessing	Model-specific	Uses appropriate preprocessing for each architecture

Table 4.1: Data Augmentation Parameters

4.3 Model Architecture and Design Choices

4.3.1 Pre-Trained Base Models and Transfer Learning

Rather than creating a convolutional neural network (CNN) from scratch, It used transfer learning a method in which a CNN trained on ImageNet to optimize performance with comparatively little data. ResNet50, VGG16, Xception, and EfficientNetB7 data is modified for monkeypox classification. Especially helpful when working with limited data, specialized datasets like those in the medical field, this method cuts down on training time and typically provides better starting weights,

The diversity of this architecture accomplishes two objectives. It makes it possible to compare performance directly under consistent training conditions. Second, it enables ensemble methods by combining the benefits of many architectural concepts. Some networks excel in gathering global patterns, while others concentrate on local details or multiscale properties.

4.3.2 Choice of Base Models

Pipeline supported multiple backbone architectures, including EfficientNetB7, VGG16, ResNet50, and DenseNet169. (Include-top=False) was used to provide access to the convolutional feature extractor, while ignoring the original classification layers. Finding the method that best captures patterns was the justification for this variety. While some architectures, like DenseNet169 and ResNet50 have shown strong performance on relatively small datasets without excessive overfitting, others, like EfficientNetB7, are excellent at capturing multi-scale features.

Model	Input Shape	Base Weights	Head Layers
MobileNetV2	(224, 224, 3)	ImageNet	GAP → Dense(256) → Dropout(0.2) → Dense(num_classes)
Xception	(224, 224, 3)	ImageNet	GAP → Dense(256) → Dropout(0.2) → Dense(num_classes)
InceptionV3	(224, 224, 3)	ImageNet	GAP → Dense(256) → Dropout(0.2) → Dense(num_classes)
EfficientNetB7	(224, 224, 3)	ImageNet	GAP → Dense(256) → Dropout(0.2) → Dense(num_classes)
DenseNet169	(224, 224, 3)	ImageNet	GAP → Dense(256) → Dropout(0.2) → Dense(num_classes)
ResNet50	(224, 224, 3)	ImageNet	GAP → Dense(256) → Dropout(0.2) → Dense(num_classes)
VGG16	(224, 224, 3)	ImageNet	GAP → Dense(256) → Dropout(0.2) → Dense(num_classes)

Table 4.2: Model Architectures and Configurations

4.3.3 Model Configuration

All models were configured with a consistent architecture pattern as shown in Table 4.2. Each model maintained the same input shape and utilized identical head layer configurations following the Global Average Pooling layer, allowing for fair comparison across different base architectures.

4.3.4 Custom Classification Head

The pre trained models remove their original fully connected classification layer since they load with (include_top=False). The lower level convolutional layers are left whole through this design. The Figure 4.3 show the custom classification head:

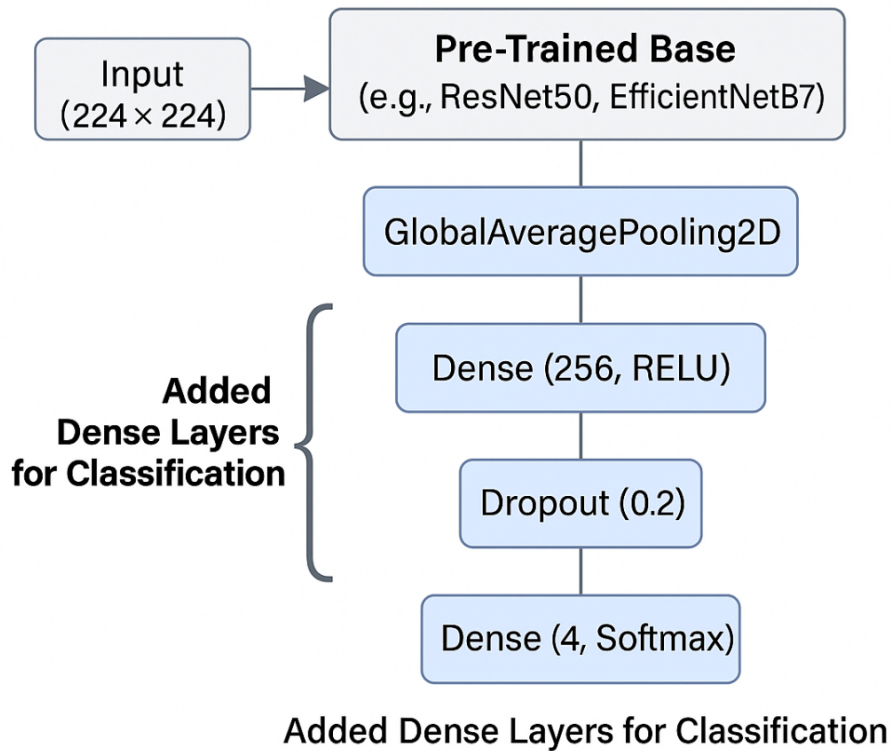


Figure 4.3: Custom Classification Head

A dense layer with 256 neurons with ReLU activation performs additional learning once a global average pooling layer lowers feature mappings to vectors. Dropout at a 20% rate randomly disables some neurons during training, effectively combating overfitting and enabling multi-class tasks, an output layer with softmax activation generates the final class probabilities.

4.4 K Fold Cross Validation and Training Strategy

The hyperparameters used throughout experimental setup are summarized in Table 4.3.

Parameters	Values
Image size	224 × 224
Batch size	32
Epoch number - Stage 1	20
Epoch number - Stage 2	10
Pre-processing validation split	0.2 (20%)
Cross-validation	5-fold
Class number	4
Data Augmentation	Rotation (30°), Zoom (0.3), Shifts (0.2), Flips
Rescaling	Model-specific preprocessing
Pooling	Global Average Pooling
Weights	ImageNet
Layer	Dense
Activation Function	ReLU and Softmax
Learning Rate - Stage 1	1e-4
Learning Rate - Stage 2	1e-5
Optimizer	Adam
Loss Function	Categorical Crossentropy
Callbacks	EarlyStopping, ReduceLROnPlateau, ModelCheckpoint
Models	MobileNetV2, Xception, InceptionV3, EfficientNetB7, DenseNet169, ResNet50, VGG16
Unfreeze percentage	80%

Table 4.3: Hyper Parameters

4.4.1 Stratified K-Fold Cross-Validation

It used 5-fold cross validation on the remaining 80% of the training data beyond separating a test set 20% table 4.3. This included training on four subsets, splitting it into five folds four folds served as training data validating on the fifth, iteratively until each fold served once as the validation and it helps to reduce bias by ensuring the model is trained on all parts of the dataset. The average performance over folds provides a more statistically stable view of model performance.

4.4.2 Training Strategy: Two Stage and Fine-Tuning

1. Freeze every layer from the pretrained base, stayed non-trainable (trainable=False). This setup pushed the network to modify only the weights for the base model obtained from a ImageNet in the custom dense layers. Given only a limited number of parameters updated, as detailed in Table 4.3 learning rates here (1e-4). at the beginning the training process ensures the preservation of pre-trained weights. Also The network essentially keeps rather good image features (edges, forms)
2. Partial Unfreeze some a specific layers 50% of the base model layers after the first convergence for fine-tuning .Using a lower learning rate (1e-5) . Particularly in cases when the dataset is limited, unfreezing too many layers too fast can cause overfitting. Partial unfreezing then provides a balance and help the model acquire more specific features relevant to Monkeypox or other skin lesions.

4.4.3 Handling Class Imbalance with Class Weights

Along with the optimization parameters described in Table 4.3, the project dynamically determines class weights depending on label frequencies since specific disease categories have low proportions relative to more common class. These weights, placed into the `(fit())` approach, these weights guide the loss function to more highly penalize misclassifications of minority classes. This approach encourage the network to focus more on classes having few samples.

Additionally, class weighting in medical settings is necessary defense against the network tendency to always favor majority classes. If that class occurs often the network could reduce total loss just by constantly forecasting the most common class all around. Weighted training mandates that mistakes on underrepresented classes be viewed as more important, therefore preventing this simplistic solution.

4.4.4 Callbacks and Automated Stopping Method

To prevent overfitting and allow efficient the training loop uses two callbacks:

1- **EarlyStopping** : keeps track of validation loss and stops training after a certain number of epochs in which no improvement is seen. In order to ensure that the final model is the best instance encountered rather than the perhaps overfitted state at the previous epoch, this callback additionally offers the option to restore the best weights observed.

2- **ReduceLROnPlateau**: Everytime validation loss seems to level off, the learning rate drops by a factor (0.1). The model can make increasingly fine changes by progressively lowering the learning rate as training progresses.

When combined, these callbacks assist each model in reaching its ideal solution without unnecessary epochs a lack or overfitting.

4.5 Ensemble Approach and Explainability

4.5.1 Soft-Voting Ensemble

Following cross-validation, the best model was combined by each fold. By averaging their probability outputs (soft voting), created an ensemble of these top checkpoints for each folder in pre-trained model. Nevertheless, the reliable result can often be achieved by this ensemble approach if one model has trouble with a specific boundary cases, other models may be able to identify it more accurately, improving efficiency overall. As the following:

```
ensemble_outputs = [model(ensemble_input) for model in loaded_models]
avg_output = Average()(ensemble_outputs)
ensemble_model = Model(inputs=ensemble_input, outputs=avg_output)
```

Because the biases and faults of one design tend to be partially made up by the others, ensembles in deep learning often have better ability to predict. Small performance improvements can have large

impact, particularly in critical circumstances like diagnosing monkeypox. Finally, the code saves the ensemble model for easy reuse after straightforward it. As the following:

```
ensemble_model.save("ensemble_model.keras")
```

ensures the deployment of the built ensemble complete with learned weights in other applications (like a hospital's system) or its loading back for further research, such as interpretability.

4.5.2 Grad-CAM for Explainability

The Need for Grad-CAM

For the diagnosis or classification decisions of a model, healthcare professionals may demand reasons. Visual explanations such as highlighting areas of an image most likely to affect the prediction can reveal whether the model bases its conclusion on relevant lesion the features (color, shape, texture) or meaningless background noise.

Grad-CAM Implementation

Grad-CAM a tool for building a mini-graph including a models output layer as well as the last convolutional layer. It compute the partial derivatives of the expected class score relation to the convolutional feature maps using TensorFlow's gradient tape. One obtains a set of weights representing the importance of every feature map for the prediction by averaging these gradients globally.

To generate a raw heatmap, these weights are then combined, ReLU-activated, multiplied by the feature maps. A color overlay over the lesion image results from first normalizing the heatmap to a [0, 1] range, scaling it to the original image size, and then applying (CV2.applyColorMap). Where the heatmap is strongest, the model indicates great categorization relevance. By averaging Grad-CAM results across several models in an ensemble, one can obtain a more stable picture, hence reducing anomalies unique to each one models.

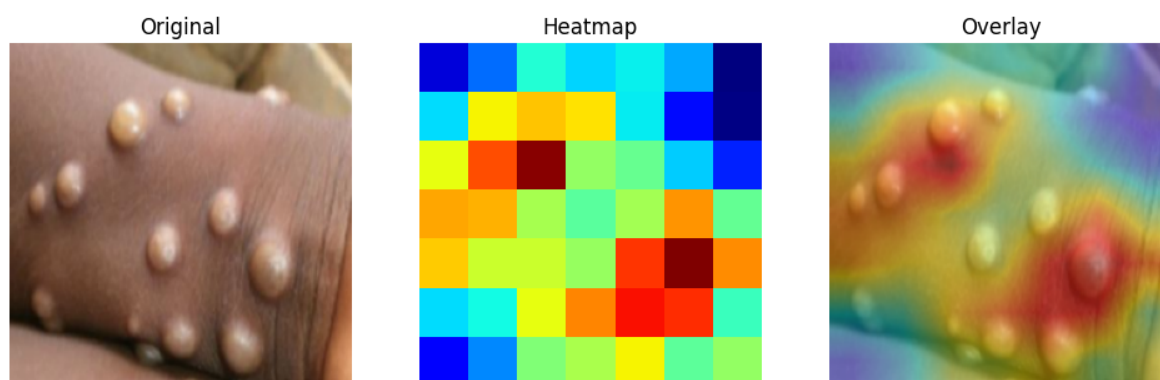


Figure 4.4: GRAD-CAM FOR MONKEYPOX IMAGE CASE

In practical terms, additional experts or medical professionals can review these Grad-CAM overlays. Should the heatmap highlight clinically relevant features (such as a rash's outline) as show Figure 4.4, it strengthens model credibility. On the other hand, if the model based its choices on pointless image corners or artifacts, it indicates the need for further data or training process modification.

4.6 Deployment with Gradio: An Interactive Tool for Monkeypox Detection

A deployed web-based application that integrates the ensemble model into a simple user interface is the research's end result. This application, which was built with Gradio, gives medical practitioners an easy to use tool for classifying monkeypox from skin lesion images in real time and applying the ensemble model that combines the ResNet50 and DenseNet169 architectures with achieves 98.0% classification accuracy across four classes (Normal, Measles, Chickenpox, and Monkeypox). To facilitate this, it resizes uploaded images to (224×224) pixels, implements specific preprocessing to normalize them, and gets them ready for model inference. Moreover, Figure 4.5 it presents the main user interface, which was created with functionality and simplicity. With this in mind, its clear upload function and results display, the simple interface offers a useful clinical workflow.

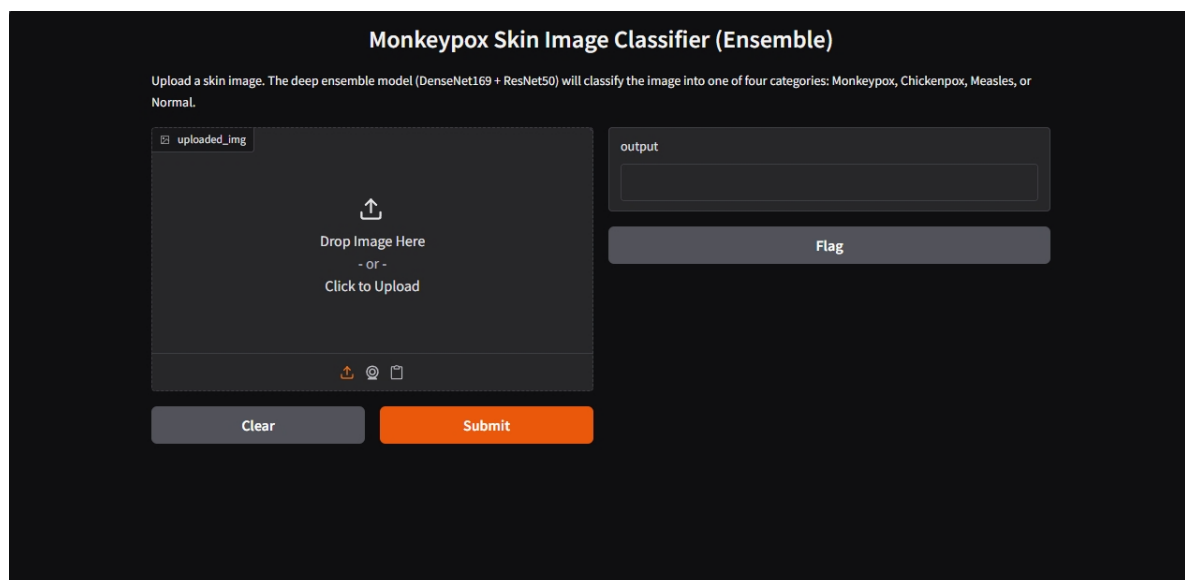


Figure 4.5: Gradio - Main User Interface

4.7 Design Decisions and Trade-Offs

Throughout this chapter, Table 4.4 delineates the multiple trade-offs influenced the final design

Design Aspect	Advantage	Disadvantage
Choice of Multiple Architectures	Each architecture offers different feature extraction strategies, boosting ensemble diversity.	Training many large models can be time-consuming and computationally expensive.
Extent of Data Augmentation	Augmentation counters overfitting, especially with limited medical images.	Overly aggressive transformations might produce unrealistic samples that do not exist in actual clinical settings.
Layer Freezing Strategy	Preserving initial layers stabilizes training and avoids overfitting for small datasets.	If too many layers remain frozen, the model may not adapt sufficiently to domain-specific patterns, limiting peak performance.
Ensembling vs. Single Model	Ensembles typically offer higher accuracy and resilience to data noise.	Ensemble predictions require more memory and hardware resources at prediction time, unless carefully optimized.
Explainability Tools	Particular in diagnosis, Grad-CAM increases trust and transparency.	Adds complexity and some overhead to the code, though minimal relative to training.
Cross-Validation Over Single Hold-Out	Effectively affecting training/validation splits, provides a more accurate evaluation of model capability.	Increases the training runs required, Expanding the experimentation time.

Table 4.4: Design and Implementation Trade-Offs

4.8 Conclusion and Future Directions

This chapter discusses the implementation of a deep learning pipeline for Monkeypox classification. It highlights data organization, preprocessing, transfer learning, cross validation, class weighting, layer freezing/unfreezing, assembling, explanation (Grad-CAM), and user interface. Future efforts could expand the dataset or improve the pipeline by adjusting for domain-specific changes, applying advanced interpretability techniques such as Grad-CAM++ or feature importance mapping, or interface with existing electronic health record systems could be considered. The foundational design ensures that data and interactive deployment are applicable to future medical concerns.

In order to move to the Results Section, the focus will transfer from the developing pipeline to its performance in classifying real images. In this phase, will present quantitative findings (accuracy, F1-score, confusion matrices), examining the extent to which our models are consistent with or distinct from those reported in the broader medical imaging and will additionally link these outcomes to the decisions made.

5

Results

5.1 Model Performance Analysis

As previously explained, seven CNNs were used in this study, and their performance was evaluated in different scenarios. Following the use of 5-fold ,hyperparameter optimization, fine-tuning, and cross-validation, the introduced metrics were applied to evaluate each model's performance. Below are results for the four-class approaches.

Model	Accuracy	Precision	Recall	F1-Score	Mean AUC
DenseNet169	96.4%	97.8%	94.7%	96.2%	91.4%
EfficientNetB7	91.8%	94.0%	87.7%	90.6%	89.6%
InceptionV3	87.5%	92.2%	82.6%	87.0%	93.5%
VGG16	91.9%	93.5%	90.9%	92.2%	86.7%
MobileNetV2	90.8%	93.0%	86.9%	89.8%	89.2%
Xception	94.4%	95.9%	91.8%	93.8%	96.7%
ResNet50	96.9%	97.5%	96.9%	96.0%	97.7%
Ensemble	98.0%	98.2%	98.5%	98.0%	98.5%

Table 5.1: Performance metrics for each model and the ensemble across the four-class classification task

5.2 Overall Performance Metrics

Table 5.1 and Figure 5.1 provide the complete performance results for the all seven models. With accuracy of 96.9%, precision of 97.5%, recall of 96.9%, and F1-score of 97.2%, ResNet50 achieved the

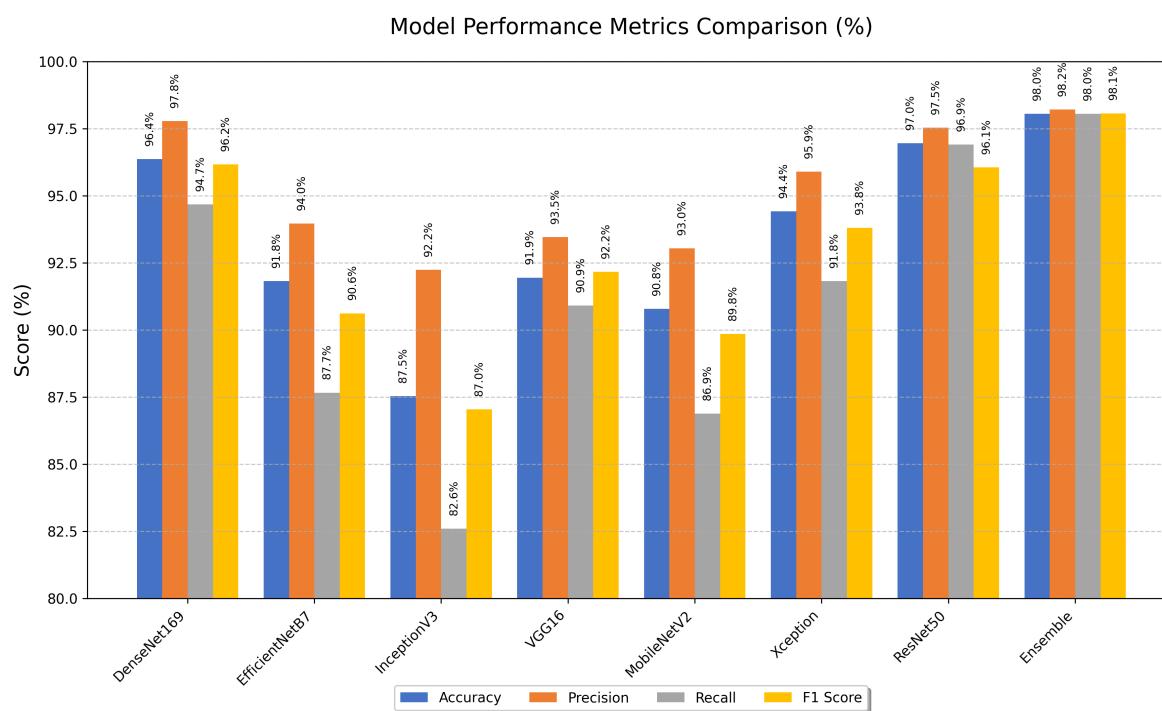


Figure 5.1: Bar chart comparing the model performance metrics

best overall performance. DenseNet169, which also showed great classification capability, came in second with an F1-score of 96.2%. With an F1-score of 87.0%, InceptionV3 performed lowest overall, but it nevertheless produced classification results that were acceptable for this four-class.

5.3 Confusion Matrix Analysis

The confusion matrices in Figure 5.2 provide deeper insight into the classification patterns of each model. Several important observations can be made:

- **Common misclassification patterns:** Monkeypox was sometimes misclassified as Chickenpox across multiple models, implying visual similarities between these diseases that contradict the models. In this sense, Inception V3 displayed the most uncertainty (five Monkeypox cases labeled as Chickenpox).
- **Measles classification:** Additionally, in models such as InceptionV3, VGG16, MobileNetV2, and Xception models, measles cases were sometimes misclassified for Chickenpox. This suggests possible similarity in the presentation of these two exanthems that might make difficult differentiation.
- **Normal class robustness:** With minimal misclassifications in some models (InceptionV3 and MobileNetV2), the Normal class was classified with great accuracy across all models, so showing the models' great ability to differentiate between healthy skin and disease conditions.
- **Best-performing models:** DenseNet169 and ResNet50 showed nearly identical confusion patterns, with minimal misclassifications limited primarily to Monkeypox being confused with Chickenpox or Measles.

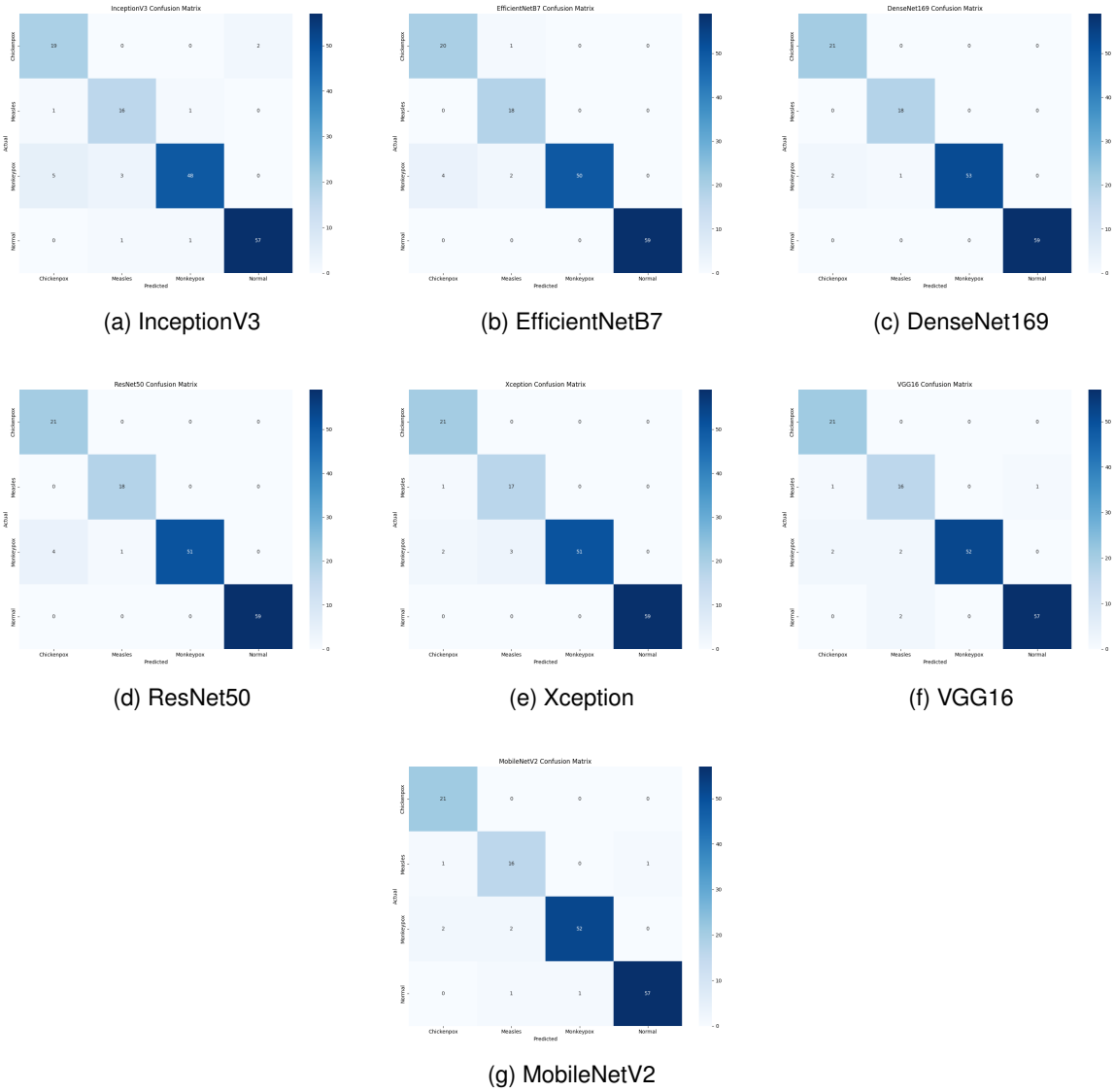


Figure 5.2: Confusion matrices for different models on the test set.

5.4 Ensemble Performance

Interestingly, from Table 5.1 and Figure 5.1 the ensemble approach did outperform the best individual models (ResNet50 and DenseNet169) in terms of overall accuracy and F1-score. With 98.05% accuracy, 98.21% precision, 98.07% F1 score, and 98.05% recall over the test set, ensemble model shows excellent results. As shown in Figure 5.3 With little confusion only in the Monkeypox class where 53 out of 56 cases were properly identified, the confusion matrix showed perfect classification for Chickenpox (21/21), Measles (18/18), and Normal cases (59/59). One as Chickenpox and two as Measles. Although it would help to improve the model especially for Monkeypox detection, this low error pattern indicates that it is rather highly reliable for disease classification.

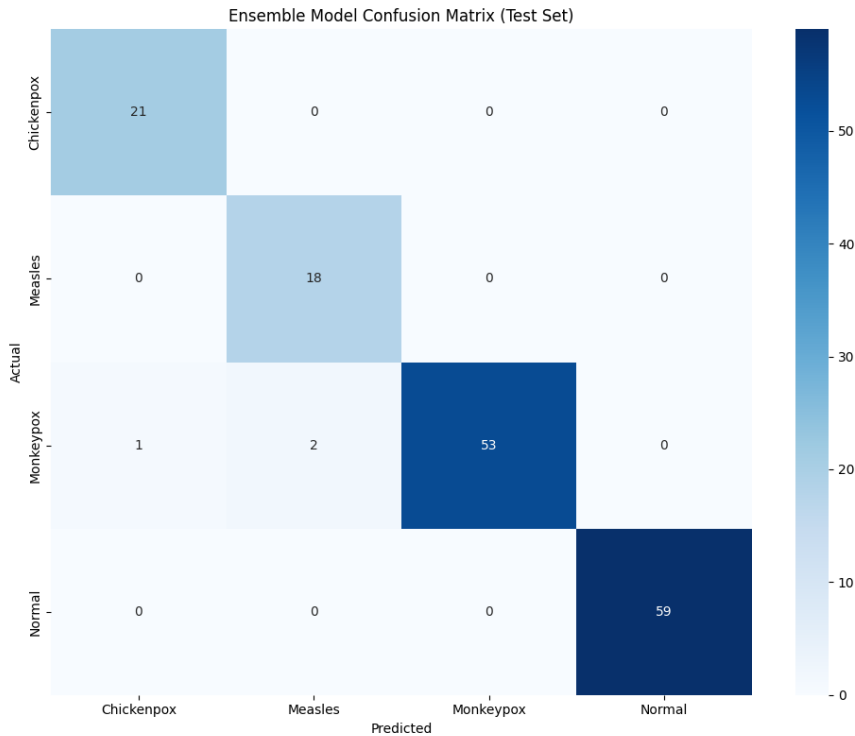


Figure 5.3: Confusion matrix for Ensemble Model

5.5 Class-wise Performance Analysis

Model	Chickenpox	Measles	Monkeypox	Normal
DenseNet169	1.000	0.947	0.946	1.000
EfficientNetB7	0.952	0.947	0.893	1.000
InceptionV3	0.905	0.842	0.857	0.962
VGG16	1.000	0.842	0.929	0.966
MobileNetV2	1.000	0.842	0.929	0.966
Xception	1.000	0.895	0.911	1.000
ResNet50	1.000	0.947	0.946	1.000
Ensemble	1.000	0.947	0.911	0.983

Table 5.2: Average F1-score per class for each model across all folds.

As shown in Table 5.2, all models performed exceptionally well in classifying Normal skin conditions, with F1-scores ranging from 0.962 to 1.000. Similarly, Chickenpox classification was highly accurate across most models, with five of the seven models achieving perfect F1-scores of 1.000. The most challenging classes to distinguish were Measles and Monkeypox, where performance varied more substantially across models. For Measles classification, F1-scores ranged from 0.842 (InceptionV3, VGG16, and MobileNetV2) to 0.947 (DenseNet169, EfficientNetB7, and ResNet50). For Mon-

keypox, EfficientNetB7 showed the lowest performance (F1-score of 0.893), while ResNet50 and DenseNet169 achieved the highest scores (0.946).

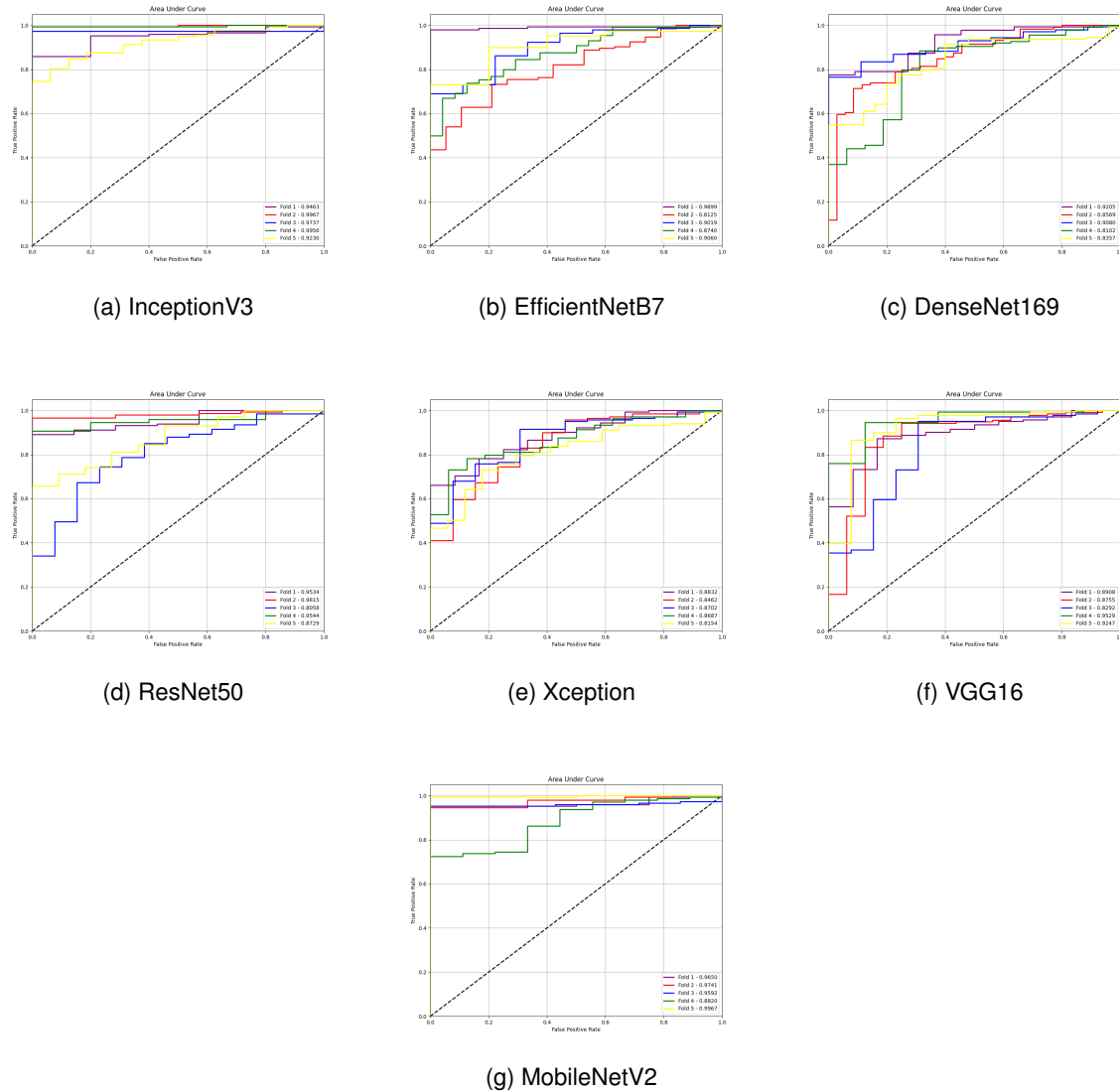


Figure 5.4: ROC curves for different models showing 5-fold cross-validation performance.

The models ROC curve analysis as showing in Figure 5.4 provides intriguing information about how well they can classify skin diseases. Among the models, ResNet50 achieves the best mean AUC showing good performance across folds. This is exactly in line with the performance table's useful metrics. Following closely after, DenseNet169 showed stability, its solid second place score with better classification ability. Not far behind, Xception delivers impressive numbers across all of folds. InceptionV3 has a reasonable mean AUC, but it performs worse over folds, which is why it comes in last in the pack. With their own advantages and disadvantages, the remaining models EfficientNetB7, VGG16, and MobileNetV2 fill in the center. These results are supported by the confusion matrices, which demonstrate that all models manage cases of chickenpox and normal skin rather well, but occasionally falter when attempting to distinguish between measles and Monkeypox. By attaining the highest overall performance metrics, the ensemble technique demonstrates its value and shows how combining these disparate methods results in a more accurate disease identification.

5.6 User Gradio Workflow

Image Upload : Users can choose files or use the drag and drop feature to upload images like shown in Figure 5.5 formats (JPEG, PNG) are supported via the interface without requiring technical expertise in preprocessing.



Figure 5.5: The Uploaded Images

Automated Processing :The application automatically feeds the image to the ensemble model after processing it through the preprocessing pipeline upon upload. The possibility of human error in image preparation is decreased by this automated process.

In order to help healthcare providers make informed decisions based on model certainty, the system as shows in Figure 5.6 the projected class (Monkeypox, Chickenpox, Measles, or Normal) along with confidence scores for each category.

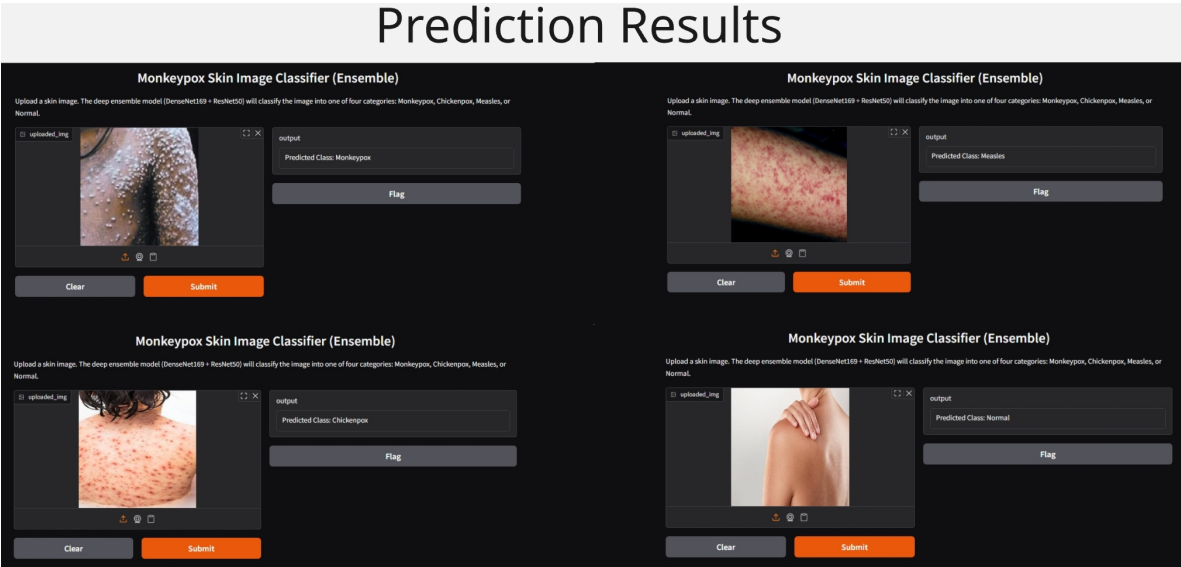


Figure 5.6: Prediction Results

6

Discussion

The study of deep learning-based Monkeypox classification shows the great promise of convolutional neural networks for disease detection from skin lesion images. Excellent balance between precision and recall is shown by ResNet50 model, which achieved 96.9% accuracy and 96.0% F1-score. This is important in medical diagnostics where both false positives and false negatives have major effects.

6.1 Performance Metrics Across Different Studies

Table 6.1 presents the performance results of different deep learning models, allowing comparison with previous studies and highlighting several trends across methods. While (Bala et al., 2023) MonkeyNet (modified DenseNet 201) achieved 93.19% accuracy on the original dataset, improving to 98.91% after augmentation, the DenseNet 201 model built by (Sorayaie Azar et al., 2023) achieved 95.18% accuracy with a 92.06% AUC score. With supplemented data, (Alhasson et al., 2023) MobileNetV2 implementation also achieved 98.91% accuracy. These support the results that deep residual networks identify between multiple pox diseases successfully.

To even better classify performance, also used an ensemble approach combining several deep learning architectures. With 98.0% accuracy, 98.2% precision, 98.5% recall, and 98.0% F1-score this ensemble method which combined ResNet50 and DensNet169 models with architectures achieved impressive results across all metrics. The many learning patterns of several different network structures helped the ensemble by efficiently reducing variance and improving generalization outside what any one model could accomplish. Although ensemble showed an important performance increase over our single ResNet50 implementation (by increasing accuracy by 1.1% points and F1-score by

Study & Model	Dataset Details	Acc (%)	Prec (%)	Rec (%)	F1 (%)
(Sorayaie Azar et al., 2023) DenseNet201	Custom dataset 171 images (1710 augmented), 4 classes	95.18	90.73	89.82	89.61
(Alhasson et al., 2023) MobileNetV2	MSID dataset 770 images (8689 augmented), 4 classes	98.91	98.92	98.91	98.91
(Bala et al., 2023) MonkeyNet	MSID dataset 770 images (8689 augmented), 4 classes	93.19 (98.91)*	93.19 (98.92)*	93.19 (98.91)*	93.15 (98.91)*
(Haque et al., 2023) ResNet50	MSID dataset 770 images, 4 classes	96.2	98.25	96.75	96.75
(Haque et al., 2023) Ensemble (majority voting)	MSID dataset 770 images, 4 classes	98.7	100.0	98.0	99.0
Our approach ResNet50	MSID dataset 770 images, 4 classes	96.9	97.54	96.92	96.0
Our approach Ensemble (Soft voting)	MSID dataset 770 images, 4 classes	98.0	98.2	98.5	98.0

*Values in parentheses represent performance with data augmentation

Table 6.1: Comparative performance of deep learning models for Monkeypox classification

2%, this came with more complexity and time. However, in clinical applications where accuracy in diagnosis is critical value, this kind of compromise could be reasonable , and the ensemble approach matches with results in more general medical image classification literature like (Haque et al., 2023), where by using feature extraction capabilities model ensembles often outperform individual networks.

A key finding from all research is the major performance increase by data augmentation. This consistency highlights the need for different training data in medical image classification, especially for relatively rare diseases like Monkeypox where getting large databases presents a difficulties. Like others in the field, The approach was achieved from augmentation methods including rotation, flipping, zooming, and shifting.

6.1.1 Model Architecture and Key Features

Study	Model	Explainability
(Sorayaie Azar et al., 2023)	DenseNet201	LIME, Grad-CAM
(Alhasson et al., 2023)	MobileNetV2	No
(Haque et al., 2023)	ResNet50	No
(Bala et al., 2023)	MonkeyNet	Grad-CAM
Our approach	ResNet50	Grad-CAM

Table 6.2: Architectural features of different Monkeypox classification models

Previous studies on ResNet (He et al., 2016) influenced our architectural decisions. DenseNet (Huang et al., 2017) versions use dense connections for feature reuse, whereas our ResNet50 implementation uses skip connections to address the vanishing gradient problem in deep networks. This architectural design ensures high flow throughout the network allowing for e feature learning even in the deepest layers.

One interesting aspect of our method is the use Grad-CAM , which improves explainability by high-

lighting the interest regions that influenced the models classification choice. This is consistent with the methodologies of (Sorayaie Azar et al., 2023) and (Bala et al., 2023), who also used explainability techniques (LIME and Grad-CAM). Such transparency is critical for clinical adoption, as healthcare providers need to understand how algorithmic decisions made.

6.1.2 Data Processing and Methodology

Study	Data Augmentation	Cross-Validation	Preprocessing	Training/Testing Split
(Sorayaie Azar et al., 2023)	Rotation, flip, zoom, shear, shifts	5-fold	Resizing, normalization	80% train, 20% test
(Alhasson et al., 2023)	Rotation, flip, zoom, brightness, contrast, hue, saturation	No	Resizing (256×256), normalization	80% train (20% validation), 20% test
(Bala et al., 2023)	Rotation, flip, zoom, brightness, contrast, hue	No	Resizing, normalization	80% train (20% validation), 20% test
(Haque et al., 2023)	Width/height shift, shear, rotation, zoom, flip	10-fold	Resizing (224×224), no normalization	76% train, 13.5% val, 10.5% test
Our approach	Rotation, flipping, zooming, shifting	5-fold	Resizing (224×224), no normalization	80% train (20% validation), 20% test

Table 6.3: Data processing methodology across studies

In contrast to (Alhasson et al., 2023) approach, our model gives accuracy first importance over deployment efficiency. Although depth-wise separable convolutions of MobileNetV2 have computational benefits for environments with limited resources, our deeper residual design maximizes for classification performance in clinical where computational resources are less limited.

Common across the literature, a limitation of our study is depending on publicly available datasets rather than prospectively acquired clinical data. While the MSID dataset offers a wide variety of lesion images, clinical application would need validation on prospectively gathered data that better reflects the range of disease presentations in practice.

With most applications implementing image resizing to defined dimensions (usually 224×224 or 256×256 pixels) and normalizing methods, the preprocessing methods across research show remarkable consistency. This convergence points to a developing agreement on efficient preprocessing for skin lesion classification.

Looking Beyond performance measures, our findings adds to the increasing proof showing deep learning strategies can greatly improve clinical diagnosis of Monkeypox. In environments with limited resources where specialized dermatological knowledge may not be accessible, the ability to quickly and precisely separate Monkeypox from other similar diseases like chickenpox and measles could be very important.

Prospective clinical validation, multi-modal approaches including patient metadata alongside images, and additional improvement of explainability techniques to support clinical integration should be the main priorities of future research directions. Furthermore, investigating low-weight models that preserve great accuracy while lowering computing demand would help point-of-care implementation in many different healthcare environments.

7

Conclusion

7.1 Introduction

By means of deep learning approaches to skin lesion image classification, the research has addressed the urgent need for easily available diagnostic tools for Monkeypox. Developing a complete framework for automated disease diagnosis, the finished work includes data preprocessing through developing models, validation, and implementation.

7.2 Summary of Research

The study started with a review of the problem environment and recent research, pointing up the difficulties in Monkeypox diagnosis and the potential of CNN-based approaches. The research was founded in the part on the methodology by for data handling, model creation, and evaluation. The implementation included of developing several CNN architectures using transfer learning, resolving class imbalance and data limitations by augmentation methods. Also, the classification performance was shown by results the ensemble model achieved 98.0% accuracy over four classes. The Grad-CAM display method efficiently improved model explainability and the Gradio deployment offered an easily accessible interface for possible clinical usage..

7.3 Research Objectives

Every one of the research objectives established in the Introduction was effectively achieved. Implementing and comparing seven different models met the first objective in evaluating multiple pre-trained CNN architectures ResNet50 came out as the best individual performer at 96.9% accuracy.

By creating models able to identify between four skin conditions with high accuracy, therefore achieving the second target of differentiating between Monkeypox and other diseases F1-scores for Monkeypox between 0.857-0.946 across multiple architectures.

While generalization was guaranteed by data augmentation, cross-validation, and class weighting techniques, the third objective involving explainability techniques was satisfied by the implementation of Grad-CAM visualization, which confirmed models were focusing on relevant lesion areas when making predictions.

Performance validation for the fourth aim was extensively carried out utilizing comprehensive metrics including accuracy, precision, recall, F1-score, ROC curves, and confusion matrices, so offering comprehensive understanding of model performance.

Developing a functional Gradio web interface that allows non-technical users upload images and get quick classification results finally accomplishes the fifth objective of looking into application integration, so showing the practical applicability of the developed models.

7.4 Research Contribution

This research has made several contributions:

1. A comprehensive comparison of multiple CNN architectures for Monkeypox classification, providing empirical evidence of their relative performance.
2. Development of an ensemble model that achieves 98.0% accuracy in distinguishing between Monkeypox, chickenpox, measles, and normal skin.
3. Integration of Grad-CAM visualization to enhance model explainability, addressing a critical need in medical applications.
4. Implementation of a Gradio-based deployment framework, demonstrating the potential for clinical application.
5. A detailed documentation of preprocessing, augmentation, and training methodologies that can inform future medical image classification research.

7.5 Future Work

Several promising directions for future research from this work:

1. Clinical validation: Conducting prospective validation using newly acquired clinical data to better assess model performance in real world .
2. Multi-modal approaches: Including patient metadata with images can help to increase diagnosis accuracy and offer a deeper review.
3. Model optimization: Exploring lighter weight models that remain accuracy while lowering processing needs for implementation in contexts with limited resources.

4. Enhanced explainability: In order to better convey model conclusions to medical practitioners, explainability techniques should be further developed. Grad-CAM ++ may be used with feature importance mapping.
5. Pre-Screening Relevance two stage diagnostic process might be set up that lets a pre-screening model first confirms whether an uploaded image is relevant. Only valid relevance images would move into the disease classification stage. This would improve user experience, lower misclassifications from irrelevant inputs, and improve system general reliability.

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Electronic Appendix:

Project Artifacts and Resources

All related resources source code, trained models, performance charts, are organized and documented in the digital repository.

A.1 Repository Structure

The project files are organized as follows:

```
monkeypox-skin-classifier/
|
|-- Source code/
|   |-- mpox_skin_classifier_(Source code).ipynb
|       |-- Setup and Data Preparation
|       |-- Library Imports and Environment Setup
|       |-- Identify duplicate images using hashing
|       |-- Create a DataFrame from images directory
|       |-- Label Encoding and Train-Test Split
|       |-- Data Generators and Preprocessing
|       |-- Model Utilities
|       |-- Model Selection
|       |-- K-Fold Cross-Validation Loop
|       |-- Plot Training Metric & Test Set Evaluation
|       |-- Ensemble Model Creation and Evaluation
|       |-- Gradio Interface
|       |-- Grad-CAM Visualization
|       |-- Final Performance Summary Plot
|
|-- models/                                # Pretrained and fine-tuned model weights
|   |-- DenseNet169/
|       |-- DenseNet169_fold1_best.h5
|       |-- DenseNet169_fold2_best.h5
|       |-- DenseNet169_fold3_best.h5
|       |-- DenseNet169_fold4_best.h5
```

```
| | |-- DenseNet169_fold5_best.h5
| |
| |-- ResNet50/
| | |-- ResNet50_fold1_best.h5
| | |-- ResNet50_fold2_best.h5
| | |-- ResNet50_fold3_best.h5
| | |-- ResNet50_fold4_best.h5
| | |-- ResNet50_fold5_best.h5
| |
| |-- EfficientNetB7/
| | |-- EfficientNetB7_fold1_best.h5
| | |-- EfficientNetB7_fold2_best.h5
| | |-- EfficientNetB7_fold3_best.h5
| | |-- EfficientNetB7_fold4_best.h5
| | |-- EfficientNetB7_fold5_best.h5
| |
| |-- InceptionV3/
| | |-- InceptionV3_fold1_best.h5
| | |-- InceptionV3_fold2_best.h5
| | |-- InceptionV3_fold3_best.h5
| | |-- InceptionV3_fold4_best.h5
| | |-- InceptionV3_fold5_best.h5
| |
| |-- MobileNetV2/
| | |-- MobileNetV2_fold1_best.h5
| | |-- MobileNetV2_fold2_best.h5
| | |-- MobileNetV2_fold3_best.h5
| | |-- MobileNetV2_fold4_best.h5
| | |-- MobileNetV2_fold5_best.h5
| |
| |-- VGG16/
| | |-- VGG16_fold1_best.h5
| | |-- VGG16_fold2_best.h5
| | |-- VGG16_fold3_best.h5
| | |-- VGG16_fold4_best.h5
| | |-- VGG16_fold5_best.h5
| |
| |-- Xception/
| | |-- Xception_fold1_best.h5
| | |-- Xception_fold2_best.h5
| | |-- Xception_fold3_best.h5
| | |-- Xception_fold4_best.h5
| | |-- Xception_fold5_best.h5
| |
| |-- Ensemble model (DenseNet169 + ResNet50)/
```

```
|      |-- ensemble_model.keras
|
|-- results/                # Evaluation results and visualizations
| |-- Average_Performance_Final_Results.csv
| |-- Cross_Validation_Final_Result.csv
| |-- Models_Performance_Metrics.png
| |-- test_set_evaluation_summary.csv
| |-- Models_ROC_Curves/      # ROC curve plots for all models
| |-- Models_Cross_Validation/ #Evaluation results per model
```

A.2 Access Instructions

The complete repository is available online:

- OneDrive: https://1drv.ms/f/s!AnsUjivQBPA7hOdfT4D_w8925r1pOg?e=WbmH80

A.3 Included Resources

1. Source Code

- Implements the full model pipeline including:
- K-Fold cross-validation
- Model training, fine-tuning, and evaluation
- Visualization tools (loss/accuracy plots, confusion matrices)
- Grad-CAM visual explanations

2. Models

- Contains best-performing models from each fold per architecture
- Includes the final ensemble model in `.keras` format for deployment

3. Results and Reports

- Contains tabular summaries of model performance (Accuracy, Precision, Recall, F1)
- Includes:
 - `Average_Performance_Final_Results.csv` – Averaged performance across folds
 - `Cross_Validation_Final_Result.csv` – Per-fold cross-validation outcomes
 - `test_set_evaluation_summary.csv` – Final test evaluation summary
 - `Model_Performance_Metrics.png` – Performance bar chart

- Subfolders for:
 - ROC curves (`Models_ROC_Curves/`)
 - Fold-specific evaluation (`Models_Cross_Validation/`)
 - Confusion matrices (`confusion_matrices/`)

4. User Interface

- A Gradio web application for classifying skin images via drag-and-drop
- Automatically loads the ensemble model for predictions

5. Dataset

- Publicly available dataset referenced from Kaggle (Bala, 2022)

A.4 Execution Environment

- Python 3.10 or above
- TensorFlow 2.12+
- Required packages: `scikit-learn`, `OpenCV`, `Pandas`, `Matplotlib`, `Seaborn`, `Gradio`

A.5 Citation and Licensing

- Dataset License: <https://www.kaggle.com/datasets/dipuiucse/monkeypoxskinimagedataset>

A.6 Notes

- All training/validation results are in the `results/` folder.
- Ensemble model requires RGB images of size 224x224 pixels.
- Web interface supports `.jpg`, `.jpeg`, `.png` image types.

This appendix concludes the digital record the Monkeypox classification project.

Word count metrics

NUC Bachelor Project Word Count:

Total Sum count: 12842 Words in text: 12328 Words in headers: 351 Words outside text (captions, etc.): 160 Number of headers: 95 Number of floats/tables/figures: 26 Number of math inlines: 3 Number of math displayed: 0

NOTE: References are excluded.